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Factors Associated with Completion of Maternity Continuum of Care in Ethiopia: A Systematic Review and Meta-Analysis

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Abstract

Objective The present study aimed to estimate the pooled prevalence of completion of maternity continuum of care and its associated factors in Ethiopia.

Design Systematic review and meta-analysis.

Study Setting Ethiopia.

Study Participants A total of 6,245 reproductive-age women were included.

Outcome measures Prevalence of completion of maternity continuum of care and factors associated with completion of maternity continuum of care.

Methods We systematically searched international databases such as MIDLINE/PubMed, Scopus, African Online Journal, Google Scholar, and Web of Sciences to retrieve related articles. The PRISMA guidelines were used for this study. Publication bias was assessed using the Funnel plot and Eggers test. Evidence of heterogeneity was checked using Cochrane Q-test and I^2 statistics. Subgroup analysis was computed based on the study region. Data were extracted using a Microsoft Excel spreadsheet and analyzed using STATA version 14 statistical software. Weighted inverse variance random effect models were used to estimate the pooled prevalence of completion of maternity continuum of care.

Results The pooled prevalence of completion of maternity continuum care in Ethiopia was 25.82% (95%CI; 16.69%-34.94%). Urban residence (AOR=2.77, 95%CI; 1.99-3.86), secondary and above educational status (AOR=3.5, 95%CI; 2.5-5) pre-pregnancy contraceptive utilization (AOR=3.25, 95%CI; 2.02-5.22), Autonomy (AOR=3.81, 95%CI; 2.74-5.31), following mass media (AOR=2.51, 95%CI; 1.79-3.50), early initiation of ANC (AOR=4.98, 95%CI; 3.28-7.57), planned pregnancy (AOR=2.93, 95%CI; 1.99-4.32), birth preparedness and complication readiness (AOR=1.8, 95%CI; 1.29-2.51), distance from a health facility (AOR=3.29, 95%CI; 2.45-4.42) were factors associated with completion of maternity continuum care in Ethiopia.

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Conclusion The pooled prevalence of completion of maternity continuum care in Ethiopia was low. Therefore, policymakers and stakeholders should give special attention to improving the completion of antenatal care, the rate of skilled birth attendant, and postnatal follow-up.

Key words: Maternity, Continuum of care, Systematic review, Meta-analysis, Ethiopia

Strength and limitation of study

- To our knowledge, this systematic review and meta-analysis is the first study conducted in Ethiopia to estimate the pooled prevalence of the maternity continuum of care and its associated factors.
- This study gives recent information on the covariates of completion of maternity continuum of care which is helpful for maternal and child health program planners.
- We had strictly followed PRISMA guidelines and assessed the quality of each study using the modified Newcastle Ottawa scale.
- The absence of study in some regions of Ethiopia might make it unable to generalize the finding of this study.
- All included studies were cross sectional which might be unable to infer a causal relationship between variables.

Background

Globally, maternal and neonatal mortality is overwhelming [1]. Approximately, 295,000 women died due to preventable pregnancy and childbirth-related complications in 2017. Of these, 94% of death occurred in low resource seating. Sub-Saharan Africa and Southern Asia accounted for the highest maternal mortality [2]. In Ethiopia, maternal mortality is 412 per

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10000 live births and neonatal mortality is 30 per 1000 live births [3]. In line with the World Health Organization, Ethiopia has planned to end preventable maternal mortality and to reduce it below 70 per 100,000 live births by the end of 2030 Sustainable Development Goals [4].

A substantial body of evidence shows that continuum of care (CoC) is one of the crucial strategies to reduce maternal and neonatal mortality in low and middle-income countries [5,6]. CoC has two components. Firstly, sequential time component including a service delivery from antenatal care, through skilled birth attendant, and postnatal follow up. Secondly, the space component includes the provision of care at the community-family level and health facility [7]. CoC has the potential of preventing half a million maternal deaths, 4 million neonatal deaths, and 6 million child death [8,9]. However, full utilization of the maternity continuum of care was a challenge in low-income countries [10]. It is estimated that only 8% to 41% of women receive a full package of CoC in Tanzania, Ghana, and Nepal [11–13].

Completion of maternity continuum care and its associated factors were studied in Ethiopia. For example, studies conducted in the Southern region reported that only 9.7% of mothers complete maternity continuum of care while in the Amhara region 12.1% to 45% of mothers completed the continuum of maternity care [10,14–20]. Evidence revealed that timely initiation of antenatal care before 16 weeks, birth preparedness and complication readiness, educational status, discussion with husband, and being satisfied with the service delivery were some of the factors associated with the completion of maternity continuum of care [15,17,18,21]. However, inconsistencies of findings concerning the completion of the maternity continuum of care and its associated factors were reported across different regions of Ethiopia. Additionally, there are no nationally representative pooled data on the prevalence of completion of maternity continuum of care in Ethiopia.

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In conclusion, understanding the overall prevalence of completion of maternity care (i.e. 4+ ANC, SBA, and at least one PNC following delivery) and the factors influencing completion of maternity continuum of care is crucial for policymakers, governments, and stakeholders to formulate strategies that enhance the continuum of maternity care and in turn, helps to improve maternal and neonatal health in the respective regions. Therefore, the main objective of this systematic review and meta-analysis was to estimate the pooled prevalence and to identify factors associated with the completion of the maternity continuum of care in Ethiopia.

Methods

Study design and setting

According to the World Bank, Ethiopia is one of the low-income countries located in the horn of Africa with a total population of 115 million, and by the end of 2060; it is projected to be 210 million. A systematic review and meta-analysis were conducted to estimate the pooled prevalence of completion of maternity continuum of care and its associated factors in Ethiopia. Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines were used for this review (Supplementary File 1) [22].

Information Sources

The PROSPERO database (http://www.library.ucsf.edu/) has been checked to rule out whether published or ongoing projects exist related to the completion of the maternity continuum of care to avoid further duplication. The finding revealed that there were no ongoing or published articles in the area of this topic. Systematic comprehensive literature was searched using international databases such as Midline/PubMed, Google Scholar, Scopus, African Online Journal, and Web of Sciences to retrieve related articles. Grey literature was searched using Google. Search terms were formulated using PICO guidelines

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through the online databases. Medical Subject Headings (MeSH) and key terms had been developed using different Boolean operators 'AND' and 'OR'. The following search terms were used: "completion" OR "Completing" OR "maternity continuum of care'' AND "Ethiopia''.

Eligibility criteria

To include studies in this systematic review and meta-analysis must meet the following criteria. Firstly, studies should report the overall prevalence of completion of maternity continuum of care and its determinants. Secondly, both published and unpublished including pre-print studies at any time in the English language only were considered. Lastly, all community-based cross-sectional study designs conducted in Ethiopia till the last search day February 18, 2022, were included. Regarding the study period, there was no restriction. Articles without full abstracts or texts and articles reported out of the outcome interest were excluded. Citations without abstracts and/or full-text, anonymous reports, commentaries, letters, editorials, reviews, and systematic reviews, Meta-analysis were excluded after reviewing the full texts.

Outcome measurements

The primary outcome was the prevalence of completion of maternity continuum care. The maternity continuum of care refers to care throughout the pregnancy, childbirth, and postnatal period. Therefore, all included studies were asked the study participants about completion of maternity continuum of care including 4+ ANC, SBA in the health facility, and PNC. Thus, if they respond "Yes" to this question they were considered as completion of maternity continuum of care if the answer was "NO'' they were considered as incomplete maternity continuum of care. The secondary outcome was factors associated with the completion of the maternity continuum of care.

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Data extraction

Initially, all relevant studies were exported to Endnote version X8 software. Duplicated studies were removed and later exported to a Microsoft Excel spreadsheet. Two authors (AM and KH) independently extracted all the important data using a standardized data extraction form which was adapted from the using the Joanna Briggs Institute (JBI) [23]. For the first outcome (prevalence) the data extraction format included (primary author, year of publication, regions, study area, sample size, and prevalence). Similarly, two authors (AM, AT) extracted data for the second outcome (associated factors of completion of maternity continuum of care) using a Microsoft excel spread sheet (i.e. adjusted odds ratio, lower confidence interval, and upper confidence interval). Finally, each factor was exported into STATA version 14 software to identify factors associated with the completion of the maternity continuum of care in Ethiopia.

Quality and risk of bias assessment

Two authors (KH and AT) were assessed the quality of each study using the modified Newcastle Ottawa Quality Assessment Scale (NOS) for cross-sectional studies was used [24]. The NOS had a maximum of 9 points. It contains sample selection, comparability, and outcome measurement; accordingly four points, two points, and three points were given respectively. The result of more than 7 points was included in this study (Supplementary File 2). In the case of disagreement between two authors; another author (AM) was involved and resolved the disagreement.

Data processing and analysis

Firstly, selected articles were entered into Microsoft Excel spread sheet format and imported to STATA version 14. statistical software for analysis. Weighted inverse variance random effect model was used to estimate the pooled prevalence of completion of maternity

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continuum of care in Ethiopia. Cochrane Q-test and I^2 statistics were computed to assess heterogeneity among all studies. Accordingly, if the result of I^2 is 0% to 40% it is mild heterogeneity, 30 to 60% would be moderate heterogeneity, 50 to 90% would be substantial heterogeneity; and 75 to 100% would be considerable heterogeneity. Funnel plot and Eggers test were done to assess publication bias. The p value >0.05 indicated that there was no publication bias. Subgroup analysis was done based on the study region. Forest plot format was used to present the pooled prevalence completion of maternity continuum of care with 95%CI. To identify factors associated with completion of maternity continuum of care, STATA version 14 Software was used.

Public and patient involvement

Neither patients nor public were involved in this study.

Results

Characteristics of included studies

The systematic searches of Midline/PubMed, Google Scholar, Scopus, African Journals Online, and Web of Sciences about the completion of maternity continuum of care and its associated factors in Ethiopia yielded 233 articles. Later on, after removing duplicated files there were 102 articles remaining. After reviewing, (n=63) articles were excluded by title, and (n=21) articles were excluded by reading abstracts. Therefore, 18 full-text articles were accessed and assessed for inclusion criteria, resulting in the further exclusion of 9 articles primarily due to reasons (Fig 1). As a result, 9 studies were fulfilled the inclusion criteria to undergo the final meta-analysis (Table 1)

Table 1. Study characteristics included in the systematic review and meta-analysis on the prevalence of completion of maternity continuum of care and its associated factors in Ethiopia.

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Study Id	Authors	Publication year	Study period	Region	Sample Size	Prevalence	Study design	NOS
1	Shitie A. et al [15]	2020	February 25 to March 10, 2019	Amhara	651	45%	cross sectional	7
2	Tizazu MA. et al [14]	2020	February 17 to March 15,2020	Amhara	647	37.2%	cross-sectional	7
3	Cherie N.et al [18]	2021	Feb-Mar 2020	Amhara	732	11.2%	cross-sectional	8
4	Asratie MH. et al [20]	2020	March 12, 2019 to April1, 2019	Amhara	819	47%	cross-sectional	8
5	Haile D. et al [17]	2020	15 February to 15 March 2019.	SNNP	432	9.7%	cross-sectional	7
6	Atnafu A. et al [19]	2020	May 01 to June 29, 2019	Amhara	565	21.6%	cross-sectional	7
7	Emiru AA. et al [10]	2020	June 2018	Amhara	1281	12.1%	cross-sectional	8
8	Sertsewold SG. et al [16]	2021	February 2020 to 10 March 2020	Amhara	614	16.1%	cross-sectional	7
9	Tsega D. et al [25]	2022	March 10 to March 30, 2020	Amhara	504	37.6%	cross-sectional	7

NB: NOS; Newcastle Ottawa Quality Assessment Scale

Completion of maternity continuum care in Ethiopia

The pooled prevalence of completion of maternity continuum of care in Ethiopia was 25.82% (95%CI; 16.69%-34.94%). The result of Cochrane heterogeneity index is (I^2=98.9%), P=0.000, indicated that there is substantial heterogeneity of different studies (I^2>50%). The finding was presented using a forest plot (Fig. 2).

Publication bias

A funnel plot was done to check the presence of publication bias at a significance level of less than 0.05. The Egger's regression test was not statistically significant (p>0.05) confirming no evidence of publication bias, as presented by the funnel plot (Fig. 3).

Subgroup analysis of completion of maternity continuum care

The finding of subgroup analysis by region showed that the pooled prevalence of completion of maternity continuum of care was highest in the Amhara region 21.53% (95% CI: 20.52-22.53), I^2 =98.9%, p <0.001) and lowest in the SNNP region 9.72% (95% CI: 6.93-12.52), I^2 = 98.9%, p <0.001) (Fig 4).

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Sensitivity analysis

A random effect model result showed that that no single study has influenced the overall pooled prevalence of completion of maternity continuum care in Ethiopia (Table 2).

Table 2, sensitivity analysis of studies included in the systematic review and meta-analysis on the prevalence of completion of maternity continuum care in Ethiopia.

Study omitted	Estimate(95% CI)
Shitie A. et al (2020)	16.07(6.07- 42.55)
Tizazu MA. et al (2020)	16.18(6.10-42.90)
Cherie N.et al (2021)	18.61(6.61-52.43)
Asratie MH. et al (2020)	15.69(5.88-41.86)
Haile D. et al (2020)	18.35(6.74-49.99
Atnafu A. et al (2020)	16.77 (6.28- 44.79)
Emiru AA. et al (2020)	19.5 (6.42-59.22)
Sertsewold SG. et al (2021)	17.18 (6.34- 46.59)
Tsega D. et al (2022)	16.42 6.24-43.24
Combined	17.07(6.65- 43.84)

Factors associated with completion of maternity continuum of care

In this systematic review and meta-analysis; urban residence, secondary and above educational status, pre-pregnancy contraceptive utilization, autonomy, following mass media, early initiation of ANC, planned pregnancy, birth preparedness, and complication readiness, distance from health facility less than 30 minutes were factors associated with completion of maternity continuum of care in Ethiopia (Supplementary File 3).

The odds of being residing in an urban area were 2.77 times more likely to complete the maternity continuum of care than those resided in a rural area (AOR=2.77, 95%CI; 1.99-3.86). Women who had secondary and above educational status were 3.5 times more likely to complete the maternity continuum of care than those who had no formal education (AOR=3.5, 95%CI; 2.5-5). The odds of utilizing pre-pregnancy contraceptives were 3.25

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times more likely to complete the maternity continuum of care than those who had not utilized contraceptive methods (AOR=3.25, 95%CI; 2.02-5.22). The odds of being autonomous in health care decisions were 3.81 times more likely to complete the maternity continuum of care than their counterparts (AOR=3.81, 95%CI; 2.74-5.31). The odds of following mass media were 2.51 times more likely to complete the maternity continuum of care than those women who did not follow mass media (AOR=2.51, 95%CI; 1.79-3.50). Women who had early initiation of ANC were 4.98 times more likely to complete the maternity continuum of care than those women who had late initiation of ANC

(AOR=4.98, 95%CI; 3.28-7.57). Women who had planned pregnancy were 2.93 times more likely to complete the maternity continuum of care than those who had unplanned pregnancy (AOR=2.93, 95%CI; 1.99-4.32). The odds of having birth preparedness and complication readiness were 1.8 times more likely to complete the maternity continuum of care than their counterparts (AOR=1.8, 95%CI; 1.29-2.51). Furthermore, women who found less than 30 minutes distance from a health facility were 3.29 times more likely to complete the maternity continuum of care than their counterparts (AOR=3.29, 95%CI; 2.45-4.42).

Discussion

The main aim of the current study was to estimate the pooled prevalence of completion of maternity continuum of care and its associated factors. Thus, the pooled prevalence of completion of maternity continuum of care was 25.82%. The finding is higher than the 2016 Ethiopian Demographic Health Survey (EDHS) (9.1%), and Ghana (8%) [12,26]. The possible explanation might be due to the study period difference. For example, our study included recent studies as compared to the 2016 EDHS report and Ghana which was conducted from 2011 to 2013. However, the finding is lower than studies conducted in Zambia (38%), Cambodia (60%) and Egypt (50.4%) of women had received the full range of

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maternity continuum of care [27–29]. The possible explanation for varieties of findings might be due to differences in accessing health care facilities. For instance, a study conducted in Cambodia might have better access to quality maternity care and full access to health care services.

Women who resided in urban areas were 2.77 times more likely to complete the maternity continuum of care than those who resided in rural areas. The possible explanation could be those women who resided in rural areas might not easily access transportation services, especially at night time and there might be a lack of access to a health facility in their nearby residence. Thus, the probability of discontinuing maternity care is more likely than those of women who resided in urban areas. The finding is in line with a study conducted in Ghana [12].

Women who had secondary and above educational status were 3.5 times more likely to complete the maternity continuum of care than those who did not have formal education. The finding is consistent with studies conducted in Pakistan, Uganda, and Kenya [30,31]. The possible explanation might be that educated women could have awareness regarding the benefit of antenatal follow up, giving birth at a health facility, and postnatal follow-up such as early detection and management of diseases, prevention of infection, and bleeding after delivery.

Completion of continuity of maternity care is higher among women who had utilized contraceptive methods before their pregnancy than those who did not utilize them. This could be explained that a family planning clinic in the health facility might be integrated with other maternal, neonatal, and child health care services. Therefore, women who had to utilize contraceptive methods have awareness concerning the continuity of care, the benefits of utilizing maternal and neonatal health care services and this might enable them to continue

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regular antenatal, childbirth, and postnatal follow-up at a health facility [32]. Women who had autonomy in health care decision-making were 3.8 times more likely to complete the maternity continuum of care than those who did not have autonomy. The finding is in line with a study conducted in Gambia [33]. This could be explained that autonomies women's have the freedom to decide on their sexual and reproductive issues when to go to health facilities and to continue their appointments. The finding is supported by a review conducted in developing countries concluded that women's autonomy is crucial to continue maternal reproductive and child health care service [34].

The odds of following mass media were 2.5 times more likely to complete the maternity continuum of care than those who did not follow mass media. The finding is similar to previous studies conducted in Zambia and Sub-Saharan Africa [28,35]. The possible explanation might be media have a significant role in creating awareness on the continuum of care, reducing knowledge gaps, and sensitizing women to have more awareness and positive attitudes towards health care-seeking behaviour.

Women who had initiated early antenatal care before 16 weeks were 4.98 times more likely to complete the maternity continuum of care than those who had initiated late ANC follow-up after 16 weeks of gestation. The possible reason might be antenatal care is the entry of maternity continuum of care, therefore, women who started early ANC visits receive counselling about iron and folic acid supplementation, birth preparedness and complication readiness, and immunization service that might enable them to complete all components of continuity care. The finding is supported by studies conducted in India, Tanzania and Ethiopia indicated that the ANC visit can enhance the continuum of care among women [13,36, 37]. A study conducted in India and Sub Saharan Africa also shows that low utilization of ANC service was a major obstacle of the continuum of care [38, 39].

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Women who had planned pregnancies were 2.93 times more likely to complete the maternity continuum of care than those who had unplanned pregnancies. This could be explained that if the pregnancy is unplanned and don't accept it by themselves, they might have fear of visiting health care facilities as compared to those who had planned pregnancy.. The finding is consistent with a study conducted in Ghana [12]. Women who had birth preparedness and complication readiness were 1.8 times more likely to complete the maternity continuum of care than their counterparts. This could be explained that those women who had birth preparedness and complication redness have awareness regarding the overall purpose of the continuum of care for their own and newborn heath. Moreover, women who found less than 30 minutes of distance from the health facility were 3.29 times more likely to complete the maternity continuum of care than their counterparts. The finding is in agreement with a study conducted in Lao PDR and Nigeria [21,40]. The possible explanation might be as the health facility is far away from their home, there might be a lack of transportation, lack of road, and might have limited information regarding the continuum of care.

Conclusion

The pooled prevalence of completion of maternity continuum of care was low in Ethiopia. Urban residence, secondary and above educational status, pre-pregnancy contraceptive utilization, autonomy, following mass media, early initiation of ANC, planned pregnancy, birth preparedness and complication readiness, distance from health facility less than 30 minutes were factors associated with completion of maternity continuum of care in Ethiopia.. Therefore, policymakers and stakeholders should give special attention to improving the completion of antenatal care, the rate of skilled birth attendant, and postnatal follow-up.

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Moreover, maternal and child health program planners should work on the identified factors to strengthen the overall continuum of maternity care in Ethiopia.

Abbreviations

ANC; Antenatal Care; CoC; Continuum of Care; SBA; Skilled Birth Attendant; PNC; Postnatal Care; NOS; Newcastle Ottawa Quality Assessment Scale; WHO; World Health Organization, PRISMA; Preferred Reporting Items for Systematic Reviews and Meta-Analyses

Declarations

Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable

Availability of data and materials

The data set analysed during the current study is available from the cross-pondering author on reasonable request.

Competing Interests

All authors declare that they have no competing interests.

Funding

Not applicable

Authors' contributions

AM conceived the idea and participated in data extraction, analysis, and draft writing. AM, KH, and AT participated in the analysis, preparation of the manuscript, and revision. All authors read and approved the final version of the manuscript to be considered for publication.

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Institutional Review Board Statement:

Not applicable. This systematic review and meta-analysis used and analysed information obtained from pre-existing studies

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Legends

Fig 1; Flow chart of selection for systematic review and meta-analysis on completion of maternity care and its associated factors in Ethiopia, 2022

Fig 2; Pooled prevalence of completion of maternity care in Ethiopia, 2022

Fig 3; Funnel plot showing symmetric distribution of articles on completion of maternity continuum of care in Ethiopia, 2022

Fig 4; forest plot showing Sub group analysis of completion of maternity continuum of care and its associated factors by region in Ethiopia, 2022

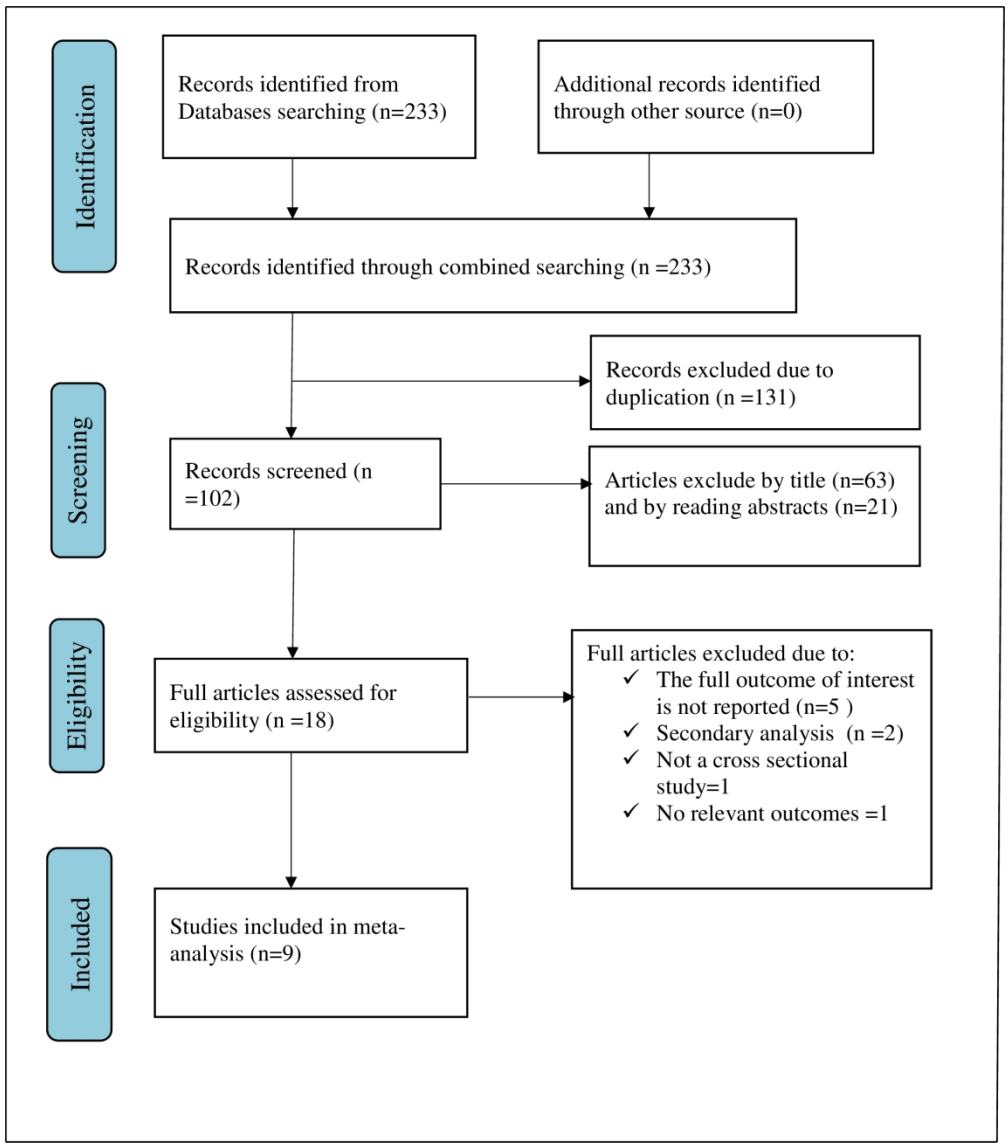
Supporting information

Supplementary File 1; Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Guideline.

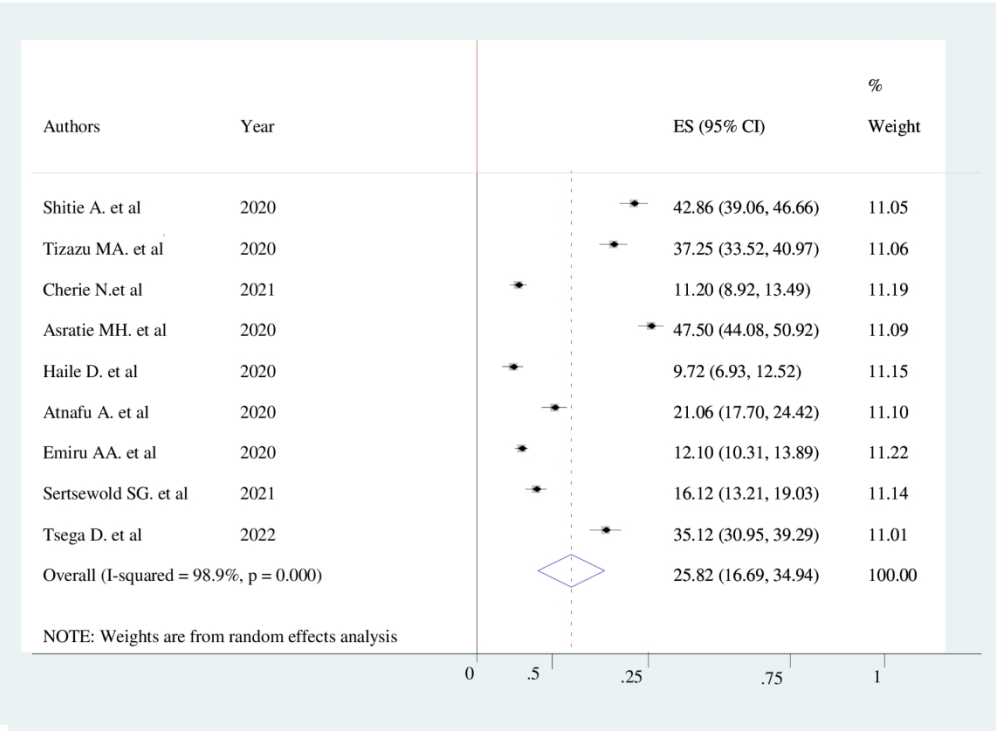
Supplementary File 2; Newcastle-Ottawa Quality Assessment Scale of 9 cross sectional studies included in the review of the status of completion of maternity continuum of care in Ethiopia

Supplementary File 3: Factors associated with completion of maternity continuum of care in Ethiopia

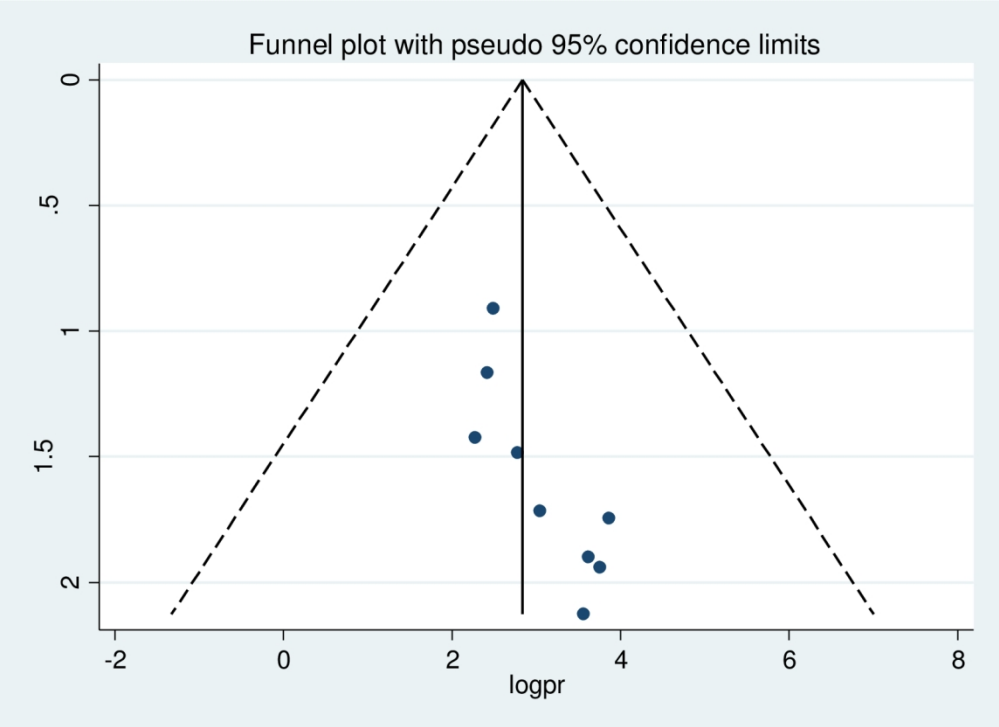
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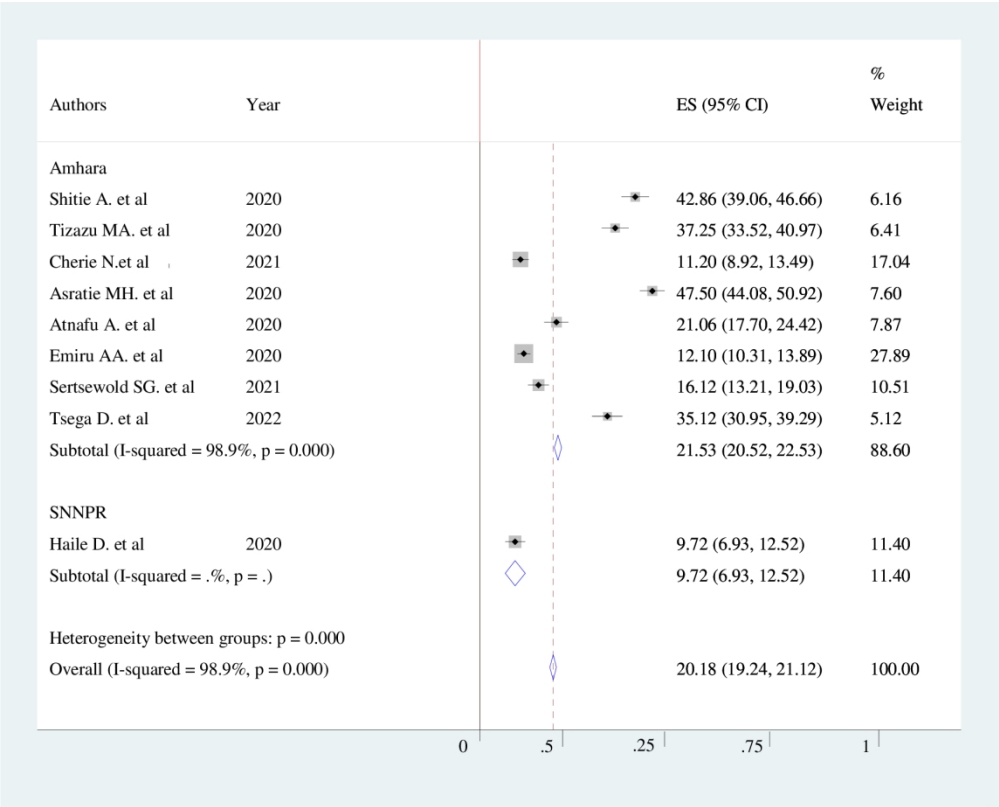
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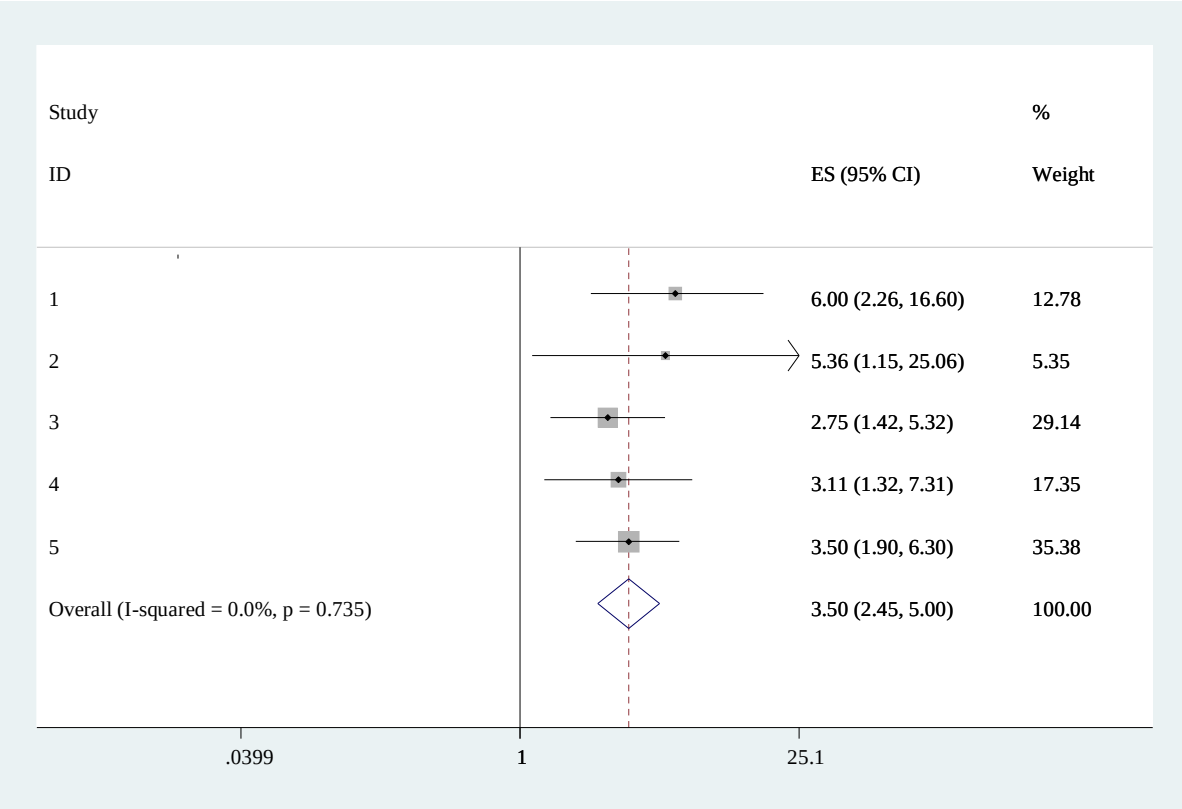
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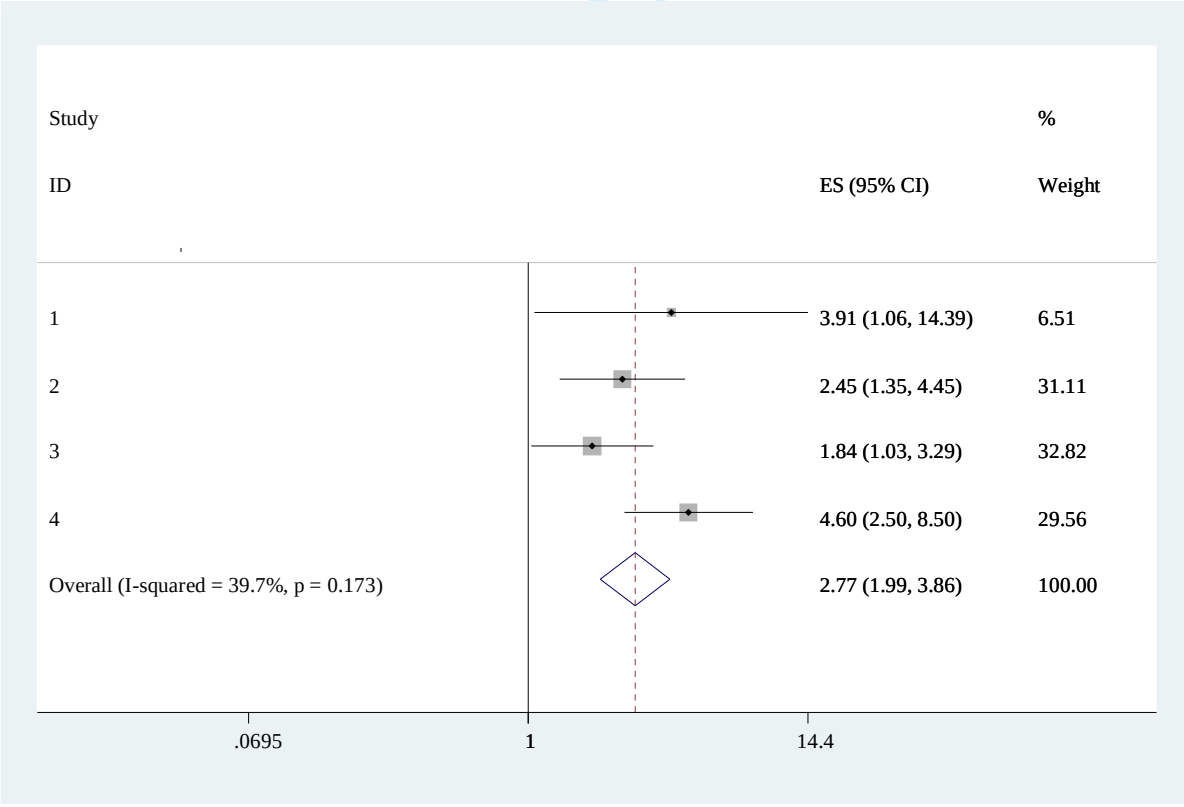
Newcastle-Ottawa Quality Assessment Scale scale for cross sectional studies	**Selection**				**Comparability**	**Outcome**		**Total score**
	Representativeness (1)	Sample size (1)	Non-respondents (1)	Ascertainment of the exposure (risk factor) (2)	The subjects in different outcome groups are comparable, based on the study design or analysis. confounding factors are controlled (2)	Assessment of the outcome (2)	Statistical test (1)	
Shitie A. et al (15)	1	1	1	1	1	1	1	7
Tizazu MA. et al (14)	1	1	1	1	1	2	1	7
Cherie N.et al(18)	1	1	1	1	1	2	1	8
Asratie MH. et al(20)	1	1	1	2	1	2	1	8
Haile D. et al(17)	1	1	1	1	1	2	1	7
Atnafu A. et al(19)	1	1	1	2	1	2	1	7
Emiru AA. et al (10)	1	1	1	2	1	2	1	8
Sertsewold SG. Et al (16)	1	1	1	1	1	2	1	7
Tsega D. et al (25)	1	1	1	1	1	1	1	7

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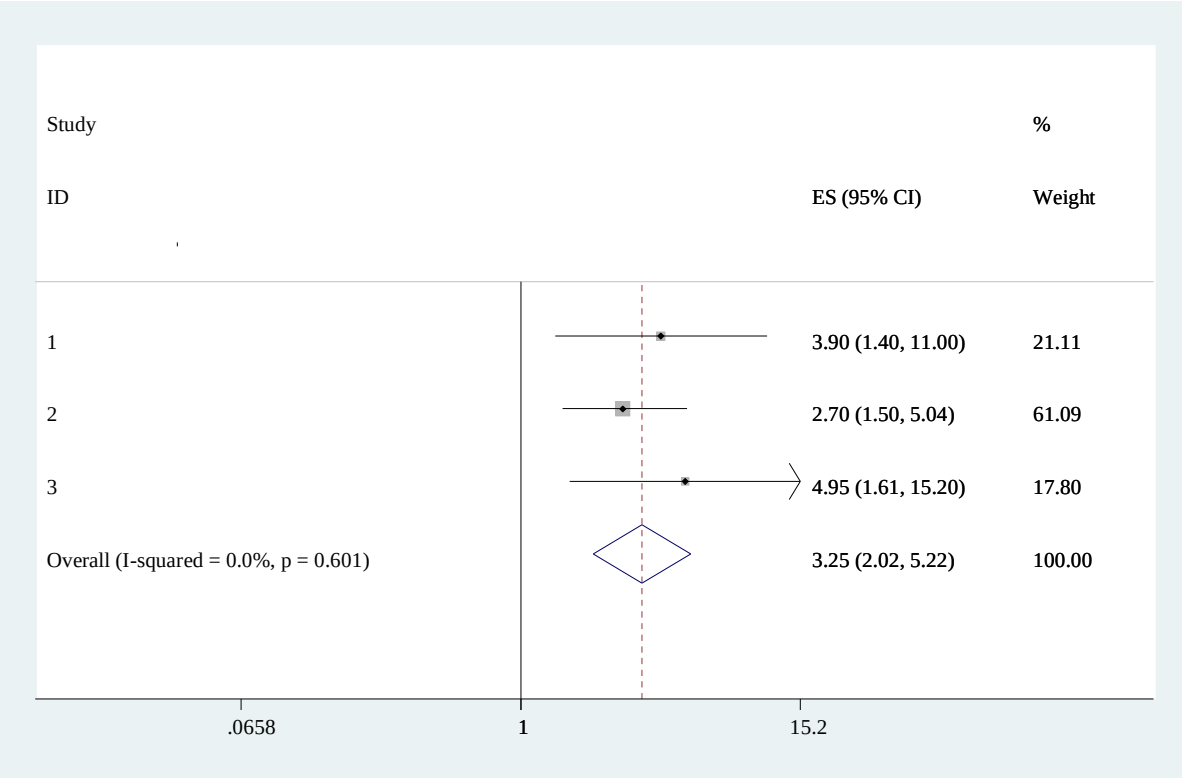
Urban residence



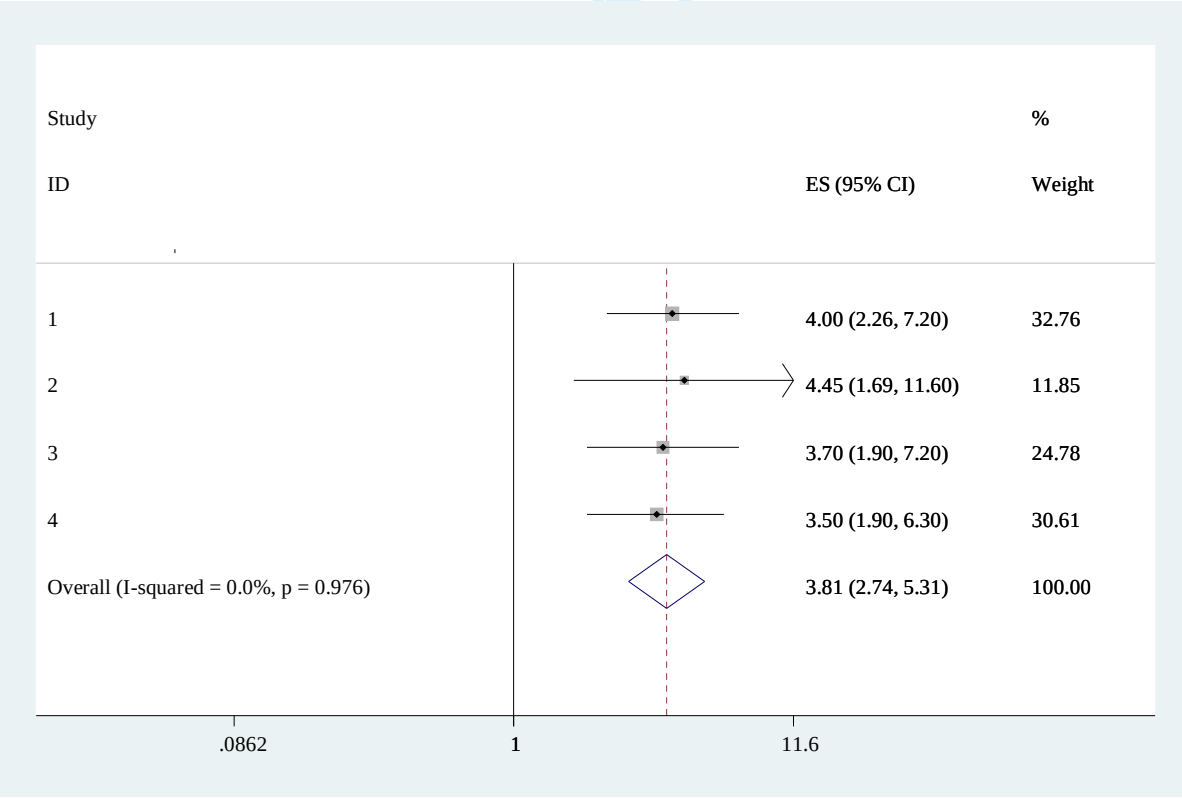
Secondary and above educational status



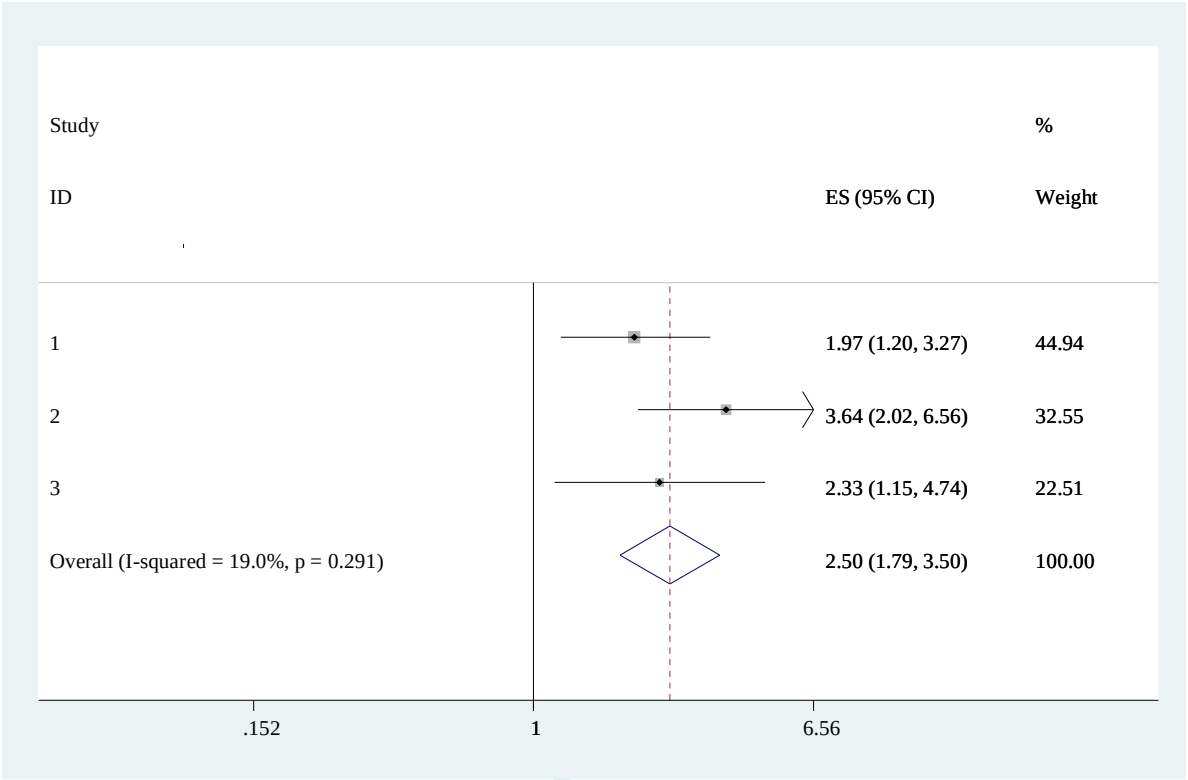
Pre-pregnancy contraceptive utilization



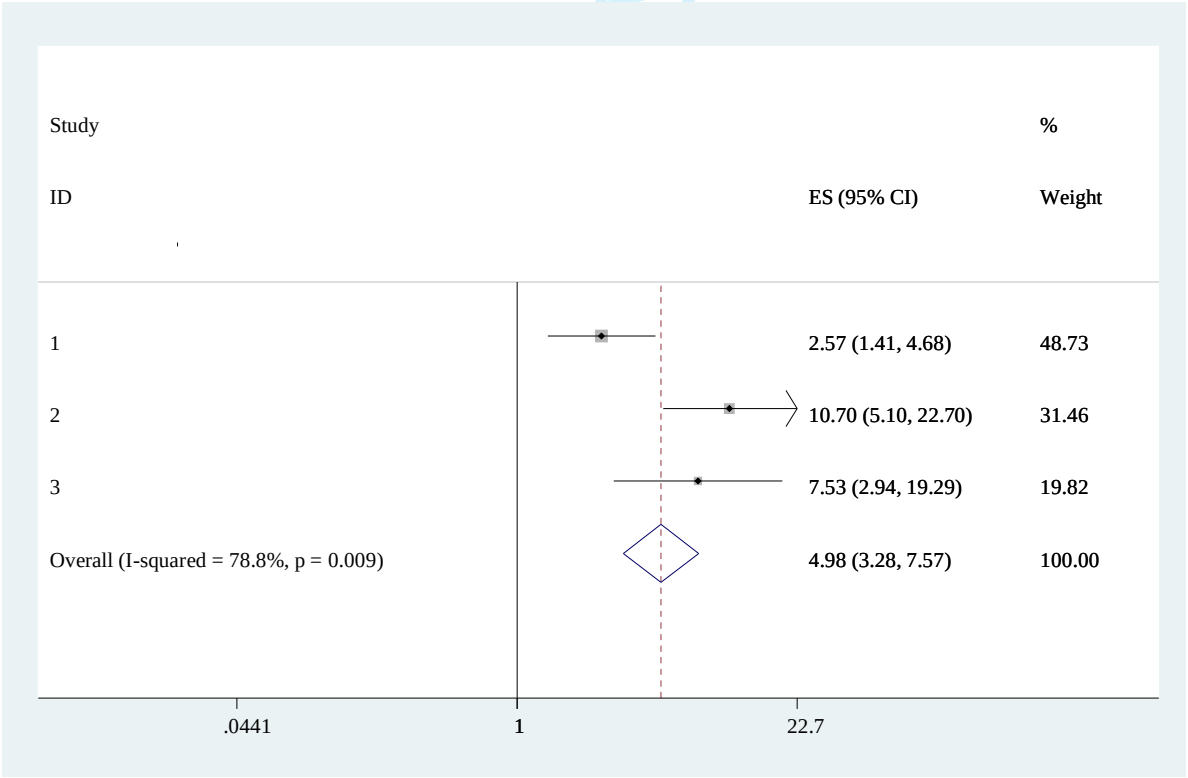
Women Autonomy



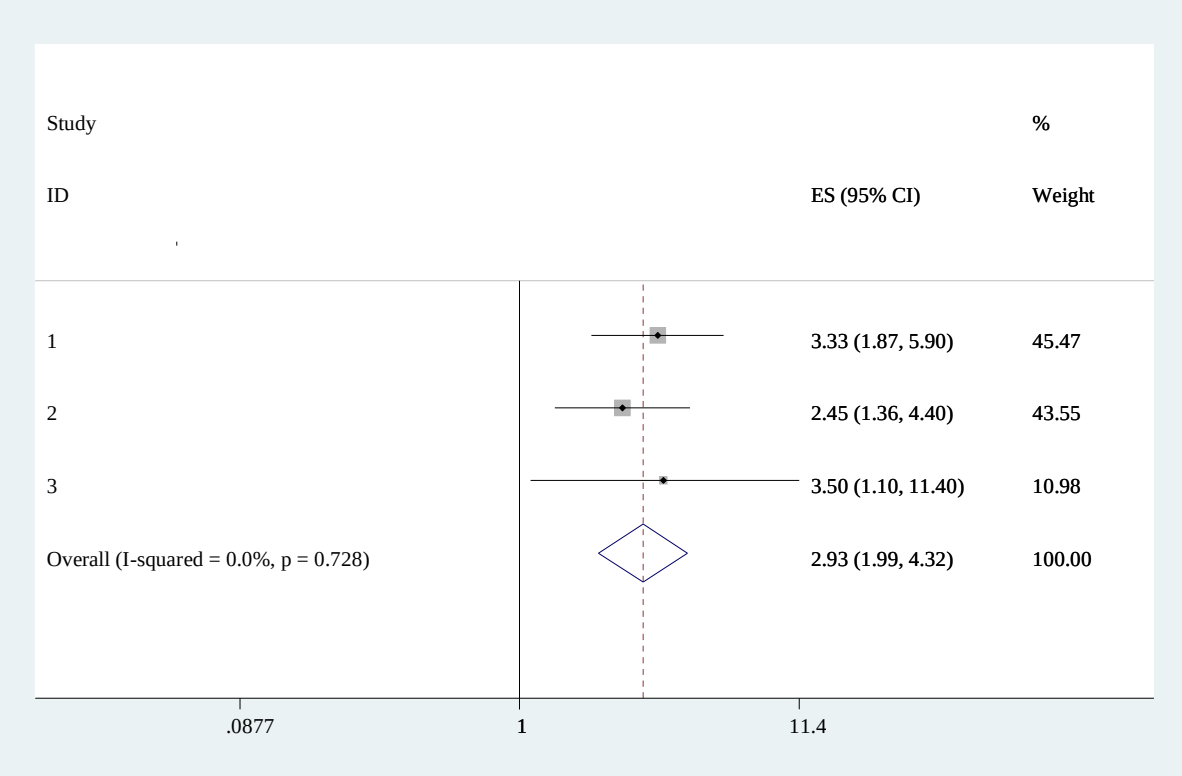
Following mass media



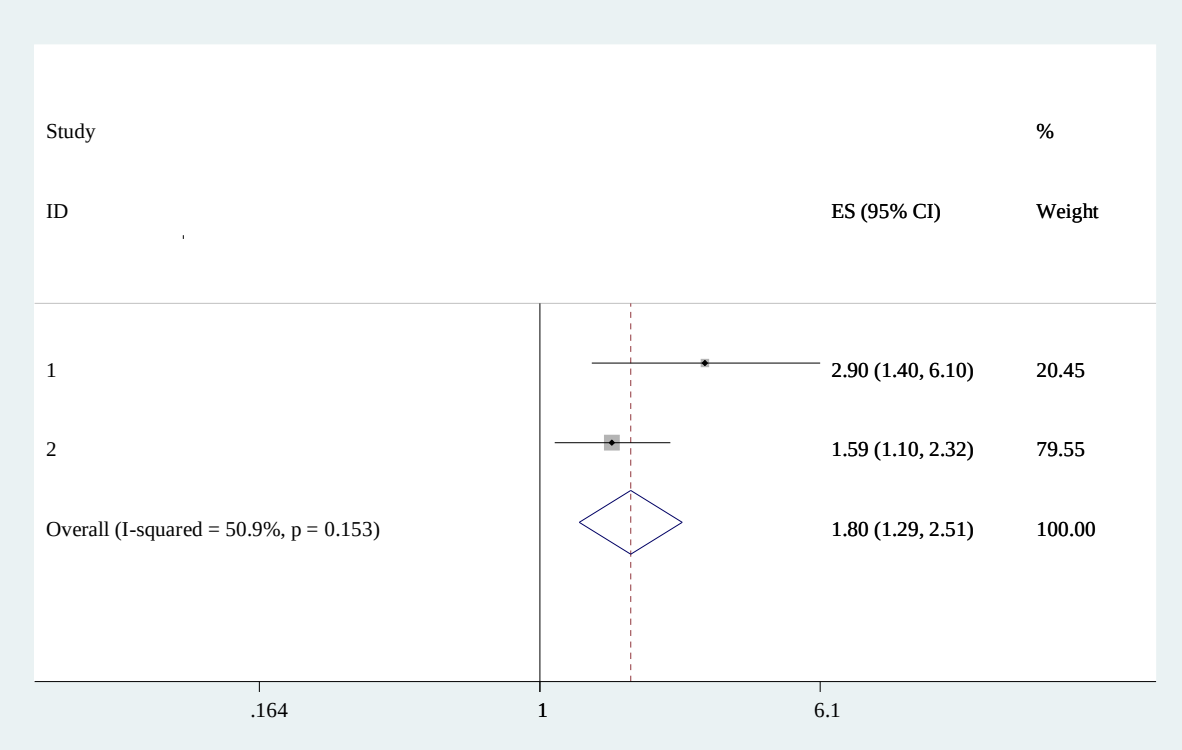
Early initiation of ANC



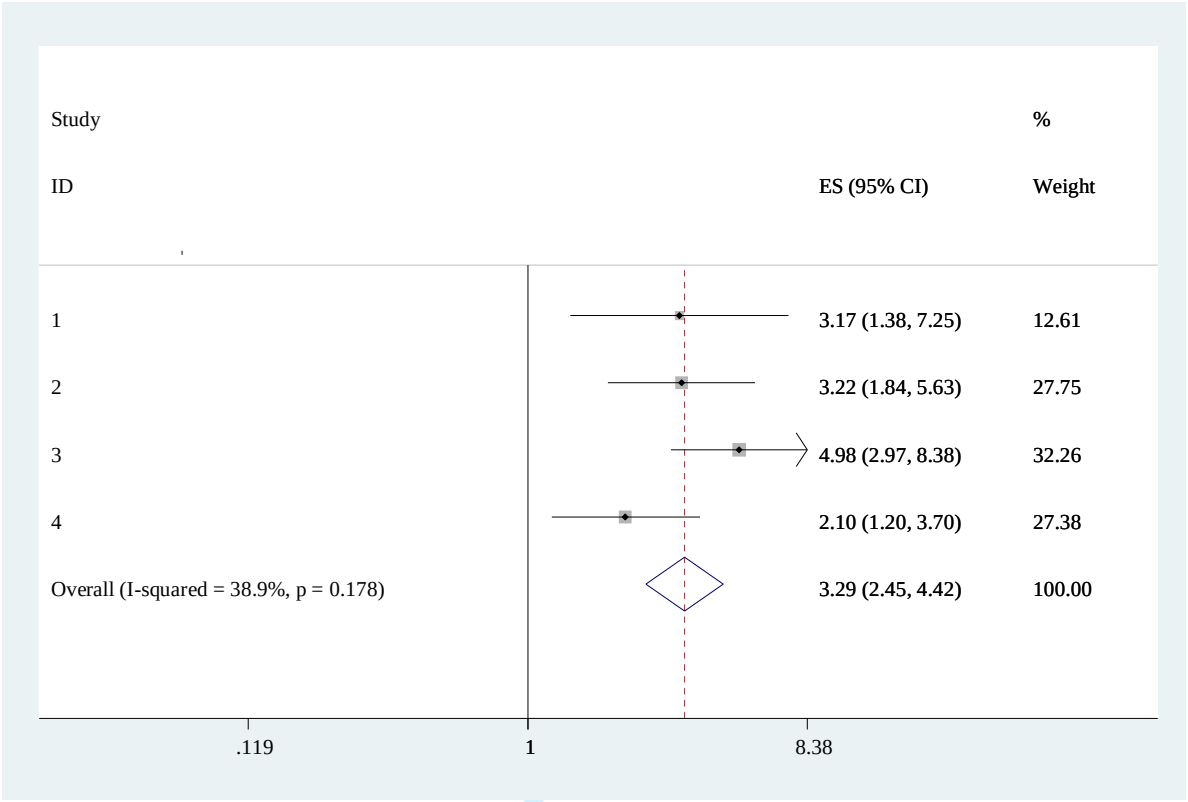
Planned pregnancy



Birth preparedness and complication readiness



Distance from health facility



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Factors Associated with Completion of Maternity Continuum of Care in Ethiopia: A Systematic Review and Meta-Analysis

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Factors Associated with Completion of Maternity Continuum of Care in Ethiopia: A Systematic Review and Meta-Analysis

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Abstract

Objective The present study aimed to estimate the pooled prevalence of completion of maternity continuum of care and its associated factors in Ethiopia.

Design Systematic review and meta-analysis.

Study Setting Ethiopia.

Study Participants A total of 6,245 reproductive-age women were included.

Outcome measures Prevalence of completion of maternity continuum of care and factors associated with completion of maternity continuum of care.

Methods We systematically searched international databases such as PubMed, Scopus, African Online Journal, Google Scholar, and Web of Sciences to retrieve related articles. The PRISMA guidelines were used for this study. Publication bias was assessed using the Funnel plot and Eggers test. Evidence of heterogeneity was checked using Cochrane Q-test and I^2 statistics. Subgroup analysis was computed based on the study region. Data were analyzed using STATA version 14 statistical software. Weighted inverse variance random effect models were used to estimate the pooled prevalence of completion of maternity continuum of care.

Results The pooled prevalence of completion of maternity continuum care in Ethiopia was 25.82% (95%CI; 16.69%-34.94%). Urban residence (AOR=2.77, 95%CI; 1.99-3.86), having secondary and above educational status (AOR=3.5, 95%CI; 2.5-5) pre-pregnancy contraceptive utilization (AOR=3.25, 95%CI; 2.02-5.22), women's autonomy (AOR=3.81, 95%CI; 2.74-5.31), following mass media (AOR=2.51, 95%CI; 1.79-3.50), early initiation of ANC (AOR=4.98, 95%CI; 3.28-7.57), planned pregnancy (AOR=2.93, 95%CI; 1.99-4.32), birth preparedness and complication readiness (AOR=1.8, 95%CI; 1.29-2.51), distance from a health facility <30 minutes (AOR=3.29, 95%CI; 2.45-4.42) were factors associated with completion of maternity continuum care in Ethiopia.

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Conclusion The pooled prevalence of completion of maternity continuum care in Ethiopia was low. Therefore, policymakers and stakeholders should give special attention to improving the completion of antenatal care, the rate of skilled birth attendants, and postnatal follow-up. Improving accessibility of health facilities, women's awareness, and empowering women's decision-making are recommended to enhance the completion of maternity continuum of care.

Key words: Maternity, Continuum of care, Systematic review, Meta-analysis, Ethiopia

Strength and limitation of study

- To our knowledge, this systematic review and meta-analysis is the first study conducted in Ethiopia to estimate the pooled prevalence of the maternity continuum of care and its associated factors.
- This study gives recent information on the covariates of completion of maternity continuum of care which is helpful for maternal and child health program planners.
- We strictly followed PRISMA guidelines and assessed the quality of each study using the modified Newcastle Ottawa scale.
- The absence of study in some regions of Ethiopia might make it unable to generalize the finding of this study.
- The difference between studies in assessing the number of antenatal contact and postnatal care (i.e. within two days or 6 weeks) might affect the outcome variables of this study. Nevertheless, each included studies were assessed each component of the maternity continuum of care.

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Background

Globally, maternal and neonatal mortality is overwhelming [1]. Approximately, 295,000 women died due to preventable pregnancy and childbirth-related complications in 2017. Of these, 94% of death occurred in low-resource seating. Sub-Saharan Africa and Southern Asia accounted for the highest maternal mortality [2]. In Ethiopia, maternal mortality is 412 per 10000 live births and neonatal mortality is 30 per 1000 live births [3]. In line with the World Health Organization, Ethiopia has planned to end preventable maternal mortality and to reduce it below 70 per 100,000 live births by the end of the 2030 Sustainable Development Goals [4].

A substantial body of evidence shows that continuum of care (CoC) is one of the crucial strategies to reduce maternal and neonatal mortality in low and middle-income countries [5,6]. CoC has two components. Firstly, the sequential time component includes a service delivery from antenatal care, through skilled birth attendants, and postnatal follow-up. Secondly, the space component includes the provision of care at the community-family level and health facility [7]. CoC has the potential of preventing half a million maternal deaths, 4 million neonatal deaths, and 6 million child death [8,9]. However, full utilization of the maternity continuum of care was a challenge in low-income countries [10]. It is estimated that only 8% to 41% of women receive a full package of CoC in Tanzania, Ghana, and Nepal [11–13].

Completion of maternity continuum care and its associated factors were studied in Ethiopia. For example, studies conducted in the Southern region reported that only 9.7% of mothers complete the maternity continuum of care while in the Amhara region 12.1% to 45% of mothers completed the continuum of maternity care [10,14–20]. Evidence revealed that timely initiation of antenatal care before 16 weeks, birth preparedness and complication

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readiness, educational status, discussion with husband, and being satisfied with the service delivery were some of the factors associated with the completion of the maternity continuum of care [15,17,18,21]. However, inconsistencies in findings concerning the completion of the maternity continuum of care and its associated factors were reported across different regions of Ethiopia. For instance, a study conducted by Shitie A. et al in the Amhara region reported that 42% of women had a complete maternity continuum of care [15]. However, a study conducted by Haile D, et al in the Southern region reported that only 9.7% of women had a complete maternity continuum of care [17]. Additionally, there are no nationally representative pooled data on the prevalence of completion of maternity continuum of care in Ethiopia.

In conclusion, understanding the overall prevalence of completion of maternity care (i.e. 4+ ANC, SBA, and at least one PNC following delivery) and the factors influencing the completion of maternity continuum of care is crucial for policymakers, governments, and stakeholders to formulate strategies that enhance the continuum of maternity care and in turn, helps to improve maternal and neonatal health in the respective regions. Therefore, the main objective of this systematic review and meta-analysis was to estimate the pooled prevalence and to identify factors associated with the completion of the maternity continuum of care in Ethiopia.

Methods

Study design and setting

According to the World Bank, Ethiopia is one of the low-income countries located in the horn of Africa with a total population of 115 million, and by the end of 2060; it is projected to be 210 million. A systematic review and meta-analysis were conducted to estimate the pooled prevalence of completion of the maternity continuum of care and its associated factors in

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Ethiopia. Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines were used for this review (Supplementary File 1) [22].

Information Sources

The PROSPERO database (http://www.library.ucsf.edu/) has been checked to rule out whether published or ongoing projects exist related to the completion of the maternity continuum of care to avoid further duplication. The finding revealed that there were no ongoing or published articles in the area of this topic. Therefore, the protocol of this review is registered at PROSPERO databases with ID=CRD42022312692. Systematic comprehensive literature was searched using international databases such as Midline/PubMed, Google Scholar, Scopus, African Online Journal, and Web of Sciences to retrieve related articles. Grey literature was searched using Google. Search terms were formulated using PICO guidelines through online databases. Medical Subject Headings (MeSH) and key terms had been developed using different Boolean operators 'AND' and 'OR' (Supplementary File 2).

Eligibility criteria

To include studies in this systematic review and meta-analysis must meet the following criteria. Firstly, studies should report the overall prevalence of completion of maternity continuum of care and its determinants. Secondly, both published and unpublished including pre-print studies at any time in the English language only were considered. Lastly, all community-based cross-sectional study designs conducted in Ethiopia till the last search day February 18, 2022, were included. Regarding the study period, there was no restriction. Articles without full abstracts or texts and articles reported out of the outcome interest were excluded. Citations without abstracts and/or full-text, anonymous reports, commentaries, letters, editorials, reviews, systematic reviews, and meta-analysis were excluded after reviewing the full texts.

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Outcome measurements

The primary outcome was the prevalence of completion of maternity continuum care. The maternity continuum of care refers to care throughout the pregnancy, childbirth, and postnatal period. In all included studies the study participants were asked about the completion of the maternity continuum of care. Firstly, having greater than or equal to 4 antenatal care visits. Antenatal care is a type of preventive healthcare which is provided in a regular check-up that allows doctors or midwives to treat and prevent potential health problems throughout the course of the pregnancy. Secondly, give birth at health facilities with a skilled birth attendant (SBA). A skilled birth attendant is a midwife, physician, and obstetrician who provide essential and emergency health care services to women and their newborns during pregnancy, and childbirth. Thirdly, receive postnatal care after delivery. It is defined as specialized care for the mother that begins immediately after delivery, up to a period of six weeks. The care includes prevention, elimination, early detection and cure of health complications, counselling on breastfeeding, and immunization. Overall, if they respond "Yes" to this question they were considered a complete maternity continuum of care if the answer was "NO'' they were considered an incomplete maternity continuum of care [7,8]. The secondary aim was to identify factors associated with the completion of the maternity continuum of care.

Data extraction

Initially, all relevant studies were exported to Endnote version X8 software. Duplicated studies were removed and later exported to a Microsoft Excel spreadsheet. Two authors (AM and KH) independently extracted all the important data using a standardized data extraction form which was adapted from the using the Joanna Briggs Institute (JBI) [23]. For the first outcome (prevalence) the data extraction format included (primary author, year of publication, regions, study area, sample size, and prevalence). Similarly, two authors (AM,

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AT) extracted data for the second outcome (associated factors of completion of maternity continuum of care) using a Microsoft Excel spreadsheet (i.e. adjusted odds ratio, lower confidence interval, and upper confidence interval). Finally, each factor was exported into STATA version 14 software to identify factors associated with the completion of the maternity continuum of care in Ethiopia.

Quality and risk of bias assessment

Three authors (AM, KH, and AT) assessed the quality of each study using the modified Newcastle Ottawa Quality Assessment Scale (NOS) for cross-sectional studies was used [24]. The NOS had a maximum of 9 points. It contains sample selection, comparability, and outcome measurement; accordingly four points, two points, and three points were given respectively. The result of more than 7 points was included in this study (Supplementary File 3). In the case of disagreement between two authors; another author (AM) was involved and resolved the disagreement.

Data processing and analysis

Firstly, selected articles were entered into Microsoft Excel spread sheet format and imported to STATA version 14. statistical software for analysis. Weighted inverse variance random effect model was used to estimate the pooled prevalence of completion of maternity continuum of care in Ethiopia. Cochrane Q-test and I^2 statistics were computed to assess heterogeneity among all studies. Accordingly, if the result of I^2 is 0% to 40% it is mild heterogeneity, 30 to 60% would be moderate heterogeneity, 50 to 90% would be substantial heterogeneity; and 75 to 100% would be considerable heterogeneity. Funnel plot and Eggers test were done to assess publication bias. The p value >0.05 indicated that there was no publication bias. Subgroup analysis was done based on the study region. Forest plot format was used to present the pooled prevalence completion of maternity continuum of care with

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95%CI. To identify factors associated with completion of maternity continuum of care, STATA version 14 Software was used.

Public and patient involvement

Neither patients nor public were involved in this study.

Results

Characteristics of included studies

The systematic searches of PubMed, Google Scholar, Scopus, African Journals Online, and Web of Sciences about the completion of maternity continuum of care and its associated factors in Ethiopia yielded 233 articles. Later on, after removing duplicated files there were 102 articles remaining. After reviewing, (n=63) articles were excluded by title, and (n=21) articles were excluded by reading abstracts. Therefore, 18 full-text articles were accessed and assessed for inclusion criteria, resulting in the further exclusion of 9 articles primarily due to reasons (Fig. 1). As a result, 9 studies were fulfilled the inclusion criteria and undergo the final meta-analysis (Table 1)

Table 1. Study characteristics included in the systematic review and meta-analysis on the prevalence of completion of maternity continuum of care and its associated factors in Ethiopia.

Study Id	Authors	Publication year	Study period	Study setting	Objective of the study	Sampling procedure	Mode of interview	Region	Sample Size	Prevalence	Study design	NOS
1	Shitie A. et al [15]	2020	February 25 to March 10, 2019	Enemay district	Assess completion Maternity Continuum of Care Identify Factors Associated with	simple random sampling technique	face-to-face interview	Amhara	651	45%	CS	7

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2	Tizazu MA. et al [14]	2020	February 17 to March 15,2020	Debre Berhan town	utilization of continuum maternity care Identify associated factors	cluster sampling technique		Amhara	647	37.2%	CS	7
3	Cherie N.et al [18]	2021	Feb-Mar 2020	Legambo district	Maternity continuum of care Determinant factors	Multistage sampling technique	face-to-face interviews	Amhara	732	11.2%	CS	8
4	Asratie MH. et al [20]	2020	March 12, 2019 to April1, 2019	Motta town and Hulet Eji Enese district	Assess completion of maternity continuum of care Identify associated factors	stratified cluster sampling technique	face to face interviews	Amhara	819	47%	CS	8
5	Haile D. et al [17]	2020	15 February to 15 March 2019.	Arba Minch Health and Demographic Surveillance System (HDSS) site	Level of completion along continuum of care Identifying associated factors	simple random sampling technique	face-to-face interviews	SNNP	432	9.7%	CS	7
6	Atnafu A. et al [19]	2020	May 01 to June 29, 2019	Dabat and Gondar Zuria rural districts	Completion of the continuum of maternal healthcare services. identify associated factors	simple random sampling technique	face-to-face interviews	Amhara	565	21.6%	CS	7
7	Emiru	2020	June 2018	West	comple	two-stage	face-to-	Amhar	1281	12.1%	CS	8

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	AA. et al [10]			Gojjam Zone	ted the continuum of maternal care services factors affecting retention on the continuum of care	cluster sampling	face interviews	a				
8	Sertsewold SG. et al [16]	2021	February 2020 to 10 March 2020	Siyadebirena Wayu district	utilisation of continuum of maternal healthcare services identify associated factors	simple random sampling	face-to-face interviews	Amhara	614	16.1%	CS	7
9	Tsega D. et al [25]	2022	March 10 to March 30, 2020	Ebinat district	maternity continuum care completion Identifying associated factors	simple random sampling technique	face-to-face interviews	Amhara	504	37.6%	CS	7

NB: NOS; Newcastle Ottawa Quality Assessment Scale; C, Cross Sectional

Completion of maternity continuum care in Ethiopia

The pooled prevalence of completion of maternity continuum of care in Ethiopia was 25.82% (95%CI; 16.69%-34.94%). The result of Cochrane heterogeneity index is (I^2=98.9%), P=0.000, indicated that there is substantial heterogeneity of different studies (I^2>50%). The finding was presented using a forest plot (Fig. 2).

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Publication bias

A funnel plot was done to check the presence of publication bias at a significance level of less than 0.05. The Egger's regression test was not statistically significant (p>0.05) confirming no evidence of publication bias, as presented by the funnel plot (Fig 3).

Subgroup analysis of completion of maternity continuum care

The finding of subgroup analysis by region showed that the pooled prevalence of completion of maternity continuum of care was highest in the Amhara region 21.53% (95% CI: 20.52-22.53), I^2 =98.9%, p <0.001) and lowest in the SNNP region 9.72% (95% CI: 6.93-12.52), I^2 = 98.9%, p <0.001) (Fig. 4).

Sensitivity analysis

A random effect model result showed that that no single study has influenced the overall pooled prevalence of completion of maternity continuum care in Ethiopia (Table 2).

Table 2, sensitivity analysis of studies included in the systematic review and meta-analysis on the prevalence of completion of maternity continuum care in Ethiopia.

Study omitted	Estimate(95% CI)
Shitie A. et al (2020)	16.07(6.07- 42.55)
Tizazu MA. et al (2020)	16.18(6.10-42.90)
Cherie N.et al (2021)	18.61(6.61-52.43)
Asratie MH. et al (2020)	15.69(5.88-41.86)
Haile D. et al (2020)	18.35(6.74-49.99
Atnafu A. et al (2020)	16.77 (6.28- 44.79)
Emiru AA. et al (2020)	19.5 (6.42-59.22)
Sertsewold SG. et al (2021)	17.18 (6.34- 46.59)
Tsega D. et al (2022)	16.42 6.24-43.24
Combined	17.07(6.65- 43.84)

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Factors associated with completion of maternity continuum of care

In this systematic review and meta-analysis; urban residence, secondary and above educational status, pre-pregnancy contraceptive utilization, autonomy, following mass media, early initiation of ANC, planned pregnancy, birth preparedness, and complication readiness, distance from health facility less than 30 minutes were factors associated with completion of maternity continuum of care in Ethiopia.

The odds of being residing in an urban area were 2.77 times more likely to complete the maternity continuum of care than those resided in a rural area (AOR=2.77, 95%CI; 1.99-3.86). Women who had secondary and above educational status were 3.5 times more likely to complete the maternity continuum of care than those who had no formal education (AOR=3.5, 95%CI; 2.5-5). The odds of utilizing pre-pregnancy contraceptives were 3.25 times more likely to complete the maternity continuum of care than those who had not utilized contraceptive methods (AOR=3.25, 95%CI; 2.02-5.22). The odds of being autonomous in health care decisions were 3.81 times more likely to complete the maternity continuum of care than their counterparts (AOR=3.81, 95%CI; 2.74-5.31). The odds of following mass media were 2.51 times more likely to complete the maternity continuum of care than those women who did not follow mass media (AOR=2.51, 95%CI; 1.79-3.50). Women who had early initiation of ANC were 4.98 times more likely to complete the maternity continuum of care than those women who had late initiation of ANC

(AOR=4.98, 95%CI; 3.28-7.57). Women who had planned pregnancy were 2.93 times more likely to complete the maternity continuum of care than those who had unplanned pregnancy (AOR=2.93, 95%CI; 1.99-4.32). The odds of having birth preparedness and complication readiness were 1.8 times more likely to complete the maternity continuum of care than their counterparts (AOR=1.8, 95%CI; 1.29-2.51). Furthermore, women who found less than 30

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minutes distance from a health facility were 3.29 times more likely to complete the maternity continuum of care than their counterparts (AOR=3.29, 95%CI; 2.45-4.42) (Table 3).

Table 3: Factors Associated with Completion of Maternity Continuum of Care in Ethiopia: A Systematic Review and Meta-Analysis, 2022.

Variables	Included studies	Odds Ratio(95%CI)	Pooled odds ratio(95%CI)	Heterogeneity (I^2 ,p-value)
Urban residence	Sertsewold SG. Et al Tsega D. et al Cherie N.et al Asratie MH. et al	3.91(1.06-14.39) 2.45(1.35-4.45) 1.84(1.03-3.29) 4.60(2.50-8.50)	2.77 (1.99, 3.86)	39.7%, 0.173
Secondary and above educational status	Shitie A. et al Sertsewold SG. Et al Tsega D. et al Tizazu MA. et al Asratie MH. et al	6.00(2.26-16.60) 5.36(1.15-25.06) 2.75(1.42-5.32) 3.11(1.32-7.31) 3.50(1.90-6.30)	3.50(2.45, 5.00)	0.0%, 0.735
Pre-pregnancy contraceptive utilization	Haile D. et al Cherie N.et al Sertsewold SG. et al	3.90(1.40-11.00) 2.70(1.50-5.04) 4.95(1.61-15.20)	3.25 (2.02,5.22)	0.0%, 0.601
Women Autonomy	Shitie A. et al Sertsewold SG. et al Cherie N.et al Asratie MH. et al	4.00(2.26-7.20) 4.45(1.69-11.60) 3.70(1.90-7.20) 3.50(1.90-6.30)	3.81 (2.74,5.31)	0.0%, 0.976
Following mass media	Shitie A. et al Tsega D. et al Cherie N.et al	1.97(1.20-3.27) 3.64(2.02-6.56) 2.33(1.15-4.74)	2.50 (1.79,3.50)	19.0%, 0.291
Early initiation of ANC	Tizazu MA. et al Haile D. et al Emiru AA. et al	2.57(1.41-4.68) 10.70(5.10-22.70) 7.53(2.94-19.29)	4.98 (3.28, 7.57)	78.8%, 0.009
Planned pregnancy	Shitie A. et al Cherie N.et al Haile D. et al	3.33(1.87-5.90) 2.45(1.36-4.40) 3.50(1.10-11.40)	2.93 (1.99,4.32)	0.0%, 0.728
Birth preparedness and complication readiness	Haile D. et al Tizazu MA. et al	2.90(1.40-6.10) 1.59(1.10-2.32)	1.80 (1.29,2.51)	50.9%, 0.153
Distance from health facility	Sertsewold SG. et al Tsega D. et al Atnafu A. et al Asratie MH. et al	3.17(1.38-7.25) 3.22(1.84-5.63) 4.98(2.97-8.38) 2.10(1.20-3.70)	3.29(2.45,4.42)	38.9%, 0.178

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Discussion

The current study revealed that the estimated prevalence of completion of maternity continuum of care was 25.82%. Urban residence, secondary and above educational status, pre-pregnancy contraceptive utilization, autonomy, following mass media, early initiation of ANC, planned pregnancy, birth preparedness and complication readiness, distance from health facility less than 30 minutes were factors associated with completion of maternity continuum of care in Ethiopia.

The pooled prevalence of completion of maternity continuum of care was 25.82%. The finding is higher than a study conducted in Ghana (8%) [12]. The possible explanation might be due to the study period difference. For example, our study included recent studies as compared to a study conducted in Ghana which was conducted from 2011 to 2013. However, the finding is lower than studies conducted in Zambia (38%), Cambodia (60%) and Egypt (50.4%) of women had received the full range of maternity continuum of care [26–28]. The possible explanation for varieties of findings might be due to differences in accessing health care facilities. For instance, a study conducted in Cambodia might have better access to quality maternity care and full access to health care services.

Women who resided in urban areas were 2.77 times more likely to complete the maternity continuum of care than those who resided in rural areas. The possible explanation could be those women who resided in rural areas might not easily access transportation services, especially at night time and there might be a lack of access to a health facility in their nearby residence. Thus, the probability of discontinuing maternity care is more likely than those of women who resided in urban areas. The finding is in line with a study conducted in Ghana [12].

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A woman's who had secondary and above educational status were 3.5 times more likely to complete the maternity continuum of care than those who did not have formal education. The finding is consistent with studies conducted in Pakistan, Uganda, and Kenya [29,30]. The possible explanation might be that educated women could have awareness regarding the benefit of antenatal follow up, giving birth at a health facility, and postnatal follow-up such as early detection and management of diseases, prevention of infection, and bleeding after delivery.

Completion of continuity of maternity care is higher among women who had utilized contraceptive methods before their pregnancy than those who did not utilize them. This could be explained that a family planning clinic in the health facility might be integrated with other maternal, neonatal, and child health care services. Therefore, women who had to utilize contraceptive methods have awareness concerning the continuity of care, the benefits of utilizing maternal and neonatal health care services and this might enable them to continue regular antenatal, childbirth, and postnatal follow-up at a health facility [31]. Women who had autonomy in health care decision-making were 3.8 times more likely to complete the maternity continuum of care than those who did not have autonomy. The finding is in line with a study conducted in Gambia [32]. This could be explained that autonomies women's have the freedom to decide on their sexual and reproductive issues when to go to health facilities and to continue their appointments. The finding is supported by a review conducted in developing countries concluded that women's autonomy is crucial to continue maternal reproductive and child health care service [33].

The odds of following mass media were 2.5 times more likely to complete the maternity continuum of care than those who did not follow mass media. The finding is similar to previous studies conducted in Zambia and Sub-Saharan Africa [27,34]. The possible explanation might be media have a significant role in creating awareness on the continuum of

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care, reducing knowledge gaps, and sensitizing women to have more awareness and positive attitudes towards health care-seeking behaviour.

Women who had initiated early antenatal care before 16 weeks were 4.98 times more likely to complete the maternity continuum of care than those who had initiated late ANC follow-up after 16 weeks of gestation. The possible reason might be antenatal care is the entry of maternity continuum of care, therefore, women who started early ANC visits receive counselling about iron and folic acid supplementation, birth preparedness and complication readiness, and immunization service that might enable them to complete all components of continuity care. The finding is supported by studies conducted in India, Tanzania and Ethiopia indicated that the ANC visit can enhance the continuum of care among women [13,35,36]. A study conducted in India and Sub Saharan Africa also shows that low utilization of ANC service was a major obstacle of the continuum of care [37, 38].

Women who had planned pregnancies were 2.93 times more likely to complete the maternity continuum of care than those who had unplanned pregnancies. This could be explained that if the pregnancy is unplanned and don't accept it by themselves, they might have fear of visiting health care facilities as compared to those who had planned pregnancy.. The finding is consistent with a study conducted in Ghana [12]. Women who had birth preparedness and complication readiness were 1.8 times more likely to complete the maternity continuum of care than their counterparts. This could be explained that those women who had birth preparedness and complication redness have awareness regarding the overall purpose of the continuum of care for their own and newborn heath. Moreover, women who found less than 30 minutes of distance from the health facility were 3.29 times more likely to complete the maternity continuum of care than their counterparts. The finding is in agreement with a study conducted in Lao PDR and Nigeria [21,39]. The possible explanation might be as the health

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facility is far away from their home, there might be a lack of transportation, lack of road, and might have limited information regarding the continuum of care.

This study has several strengths. To our knowledge, this systematic review and meta-analysis is the first study conducted in Ethiopia to estimate the pooled prevalence of the complete maternity continuum of care and its associated factors. Additionally, this study gives recent information on the covariates of completion of the maternity continuum of care which is helpful for maternal and child health program planners. Moreover, we strictly followed PRISMA guidelines and assessed the quality of each study using the modified Newcastle Ottawa scale. However, this study is not void of limitations. Differences between studies in assessing the number of antenatal contact and postnatal care within two days or 6 weeks might affect the outcome variables of this study. Moreover, the absence of study in some regions of Ethiopia (i.e. Afar, Gambella, Somalia, and Tigray region) might make it unable to generalize the finding of this study. Thus, we recommend that future studies should be conducted in the above respective regions.

Conclusion

The pooled prevalence of completion of the maternity continuum of care was low in Ethiopia. Urban residence, secondary and above educational status, pre-pregnancy contraceptive utilization, women's autonomy, following mass media, early initiation of ANC, planned pregnancy, birth preparedness, and complication readiness, and distance from health facility less than 30 minutes were factors associated with completion of maternity continuum of care in Ethiopia. Therefore, policymakers and stakeholders should give special attention to improving the completion of antenatal care, the rate of skilled birth attendants, and postnatal follow-up. Moreover, maternal and child health program planners should work on the identified factors to strengthen the overall continuum of maternity care in Ethiopia.

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Abbreviations

ANC; Antenatal Care; CoC; Continuum of Care; SBA; Skilled Birth Attendant; PNC; Postnatal Care; NOS; Newcastle Ottawa Quality Assessment Scale; WHO; World Health Organization, PRISMA; Preferred Reporting Items for Systematic Reviews and Meta-Analyses

Declarations

Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable

Availability of data and materials

The data set analysed during the current study is available from the cross-pondering author on reasonable request.

Competing Interests

All authors declare that they have no competing interests.

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Authors' contributions

AM conceived the idea and participated in data extraction, analysis, and draft writing. AM, KH, and AT participated in the analysis, preparation of the manuscript, and revision. All authors read and approved the final version of the manuscript to be considered for publication.

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Institutional Review Board Statement:

Not applicable. This systematic review and meta-analysis used and analysed information obtained from pre-existing studies

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doi:10.1186/s12889-016-3075-0

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Legends

Fig 1; Flow chart of selection for systematic review and meta-analysis on completion of maternity care and its associated factors in Ethiopia, 2022

Fig 2; Pooled prevalence of completion of maternity care in Ethiopia, 2022

Fig 3; Funnel plot showing symmetric distribution of articles on completion of maternity continuum of care in Ethiopia, 2022

Fig 4; forest plot showing Sub group analysis of completion of maternity continuum of care and its associated factors by region in Ethiopia, 2022

Supporting information

Supplementary File 1; Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Guideline.

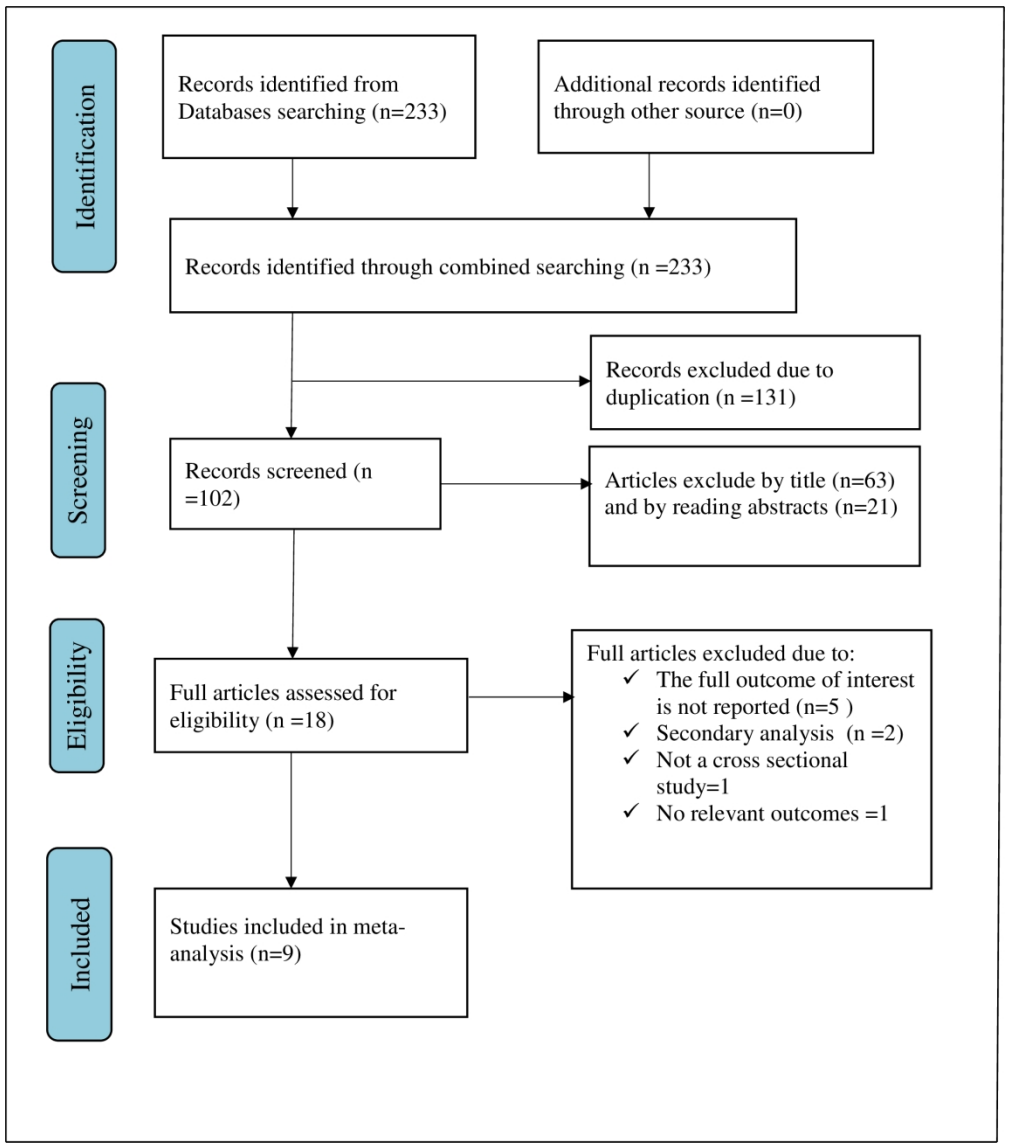
Supplementary File 2: Search string for databases.

Supplementary File 3; Newcastle-Ottawa Quality Assessment Scale of 9 cross sectional studies included in the review of the status of completion of maternity continuum of care in Ethiopia

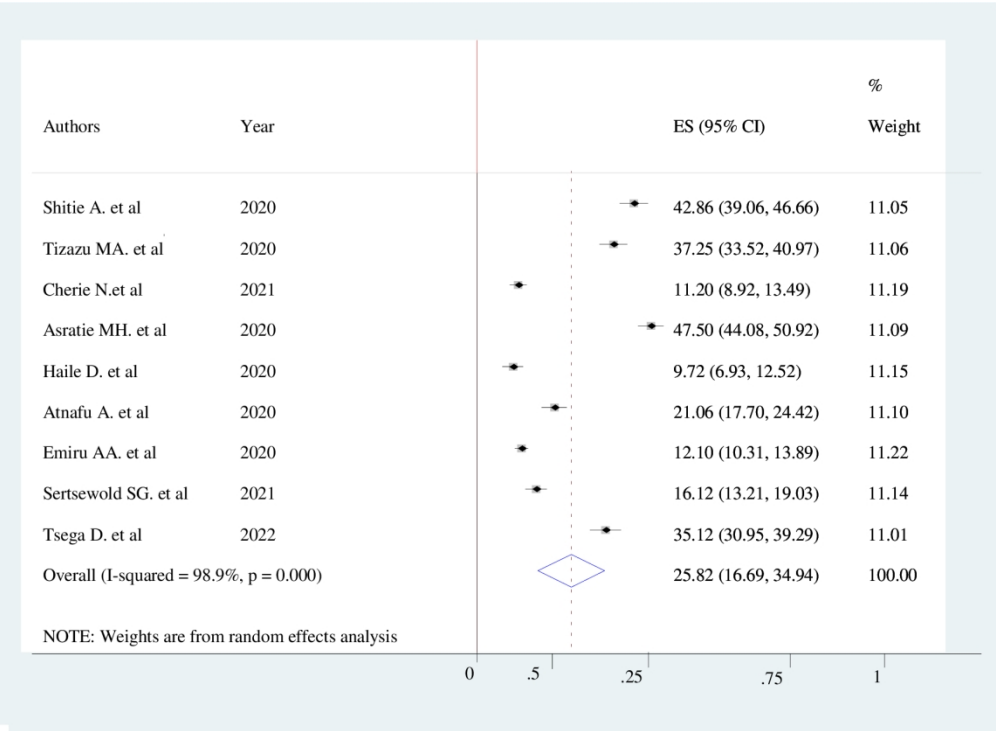
: Factors associated with completion of maternity continuum of care in Ethiopia

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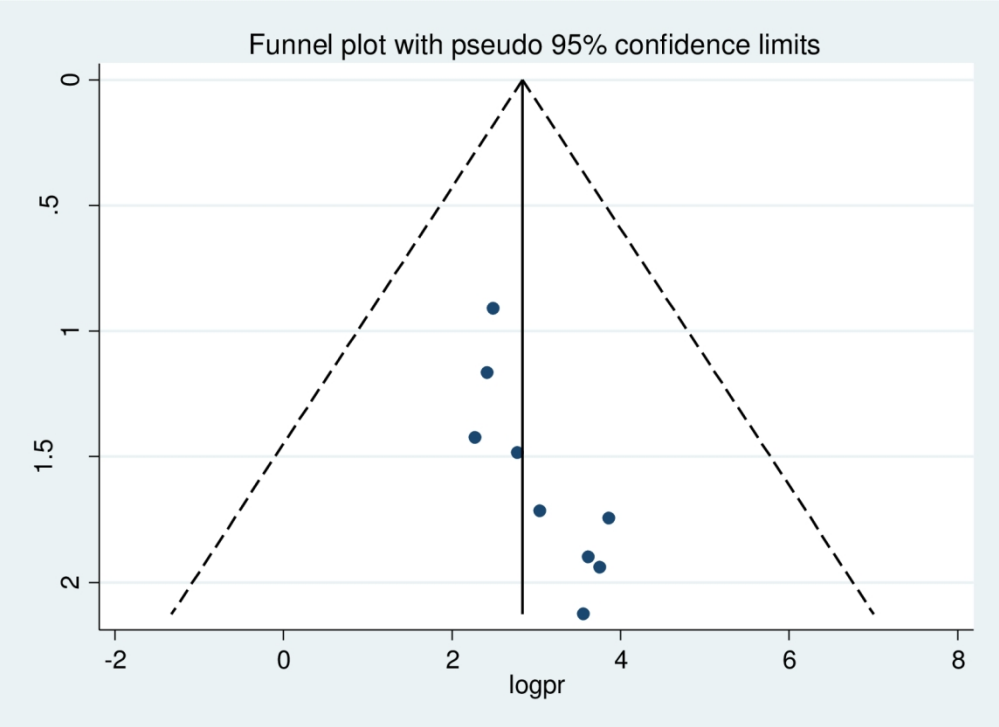
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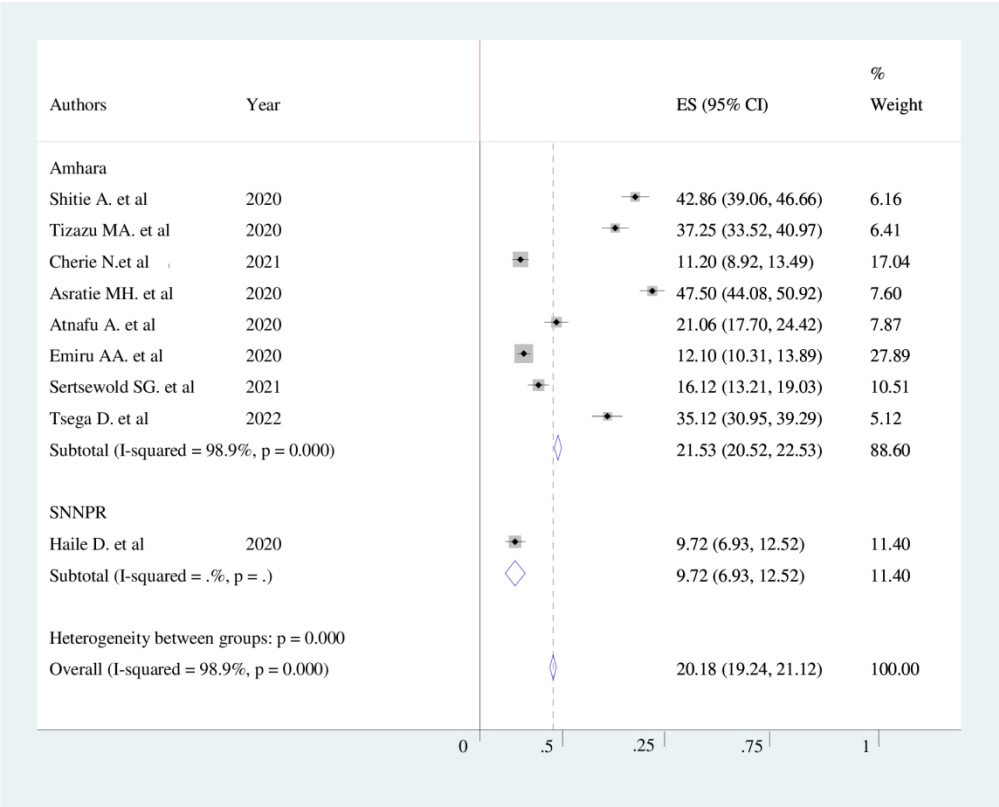
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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2 to 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 3
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 5 to 6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 7
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 7
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 6
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 7
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 7
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 7
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 9
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 9
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 9
Study characteristics	17	Cite each included study and present its characteristics.	Page 8
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 8 to page 13
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Page 8 to page 13
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page 8 to page 13
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Page 8 to page 13
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 8 to page 13
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Page 8 to page 13
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 8 to page 13
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 8 to page 13
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 14 to 17
	23b	Discuss any limitations of the evidence included in the review.	Page 18
	23c	Discuss any limitations of the review processes used.	Page1 8
	23d	Discuss implications of the results for practice, policy, and future research.	Page 18
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 5 to 6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 5
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 13
Competing interests	26	Declare any competing interests of review authors.	Page 12
Availability of data, code and	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 18 to 19

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
other materials			

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

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Databases	Search	Search terms
PubMed	#1	"Completion" OR "Completing"
	#2	"Maternity Continuum of Care"
	#3	"Completion" OR "Completing" AND "maternity continuum of care"
	#4	"Associated factors" OR "Determinates factors" OR "Predictors"
	#5	"Postpartum Mothers"
	#6	"Completion" OR "Completing" AND "maternity continuum of care" AND "Associated factors" OR "Determinates factors" OR "Predictors" AND "Postpartum Mothers"
	#7	"Completion" OR "Completing" AND "maternity continuum of care" AND "Associated factors" OR "Determinates factors" OR "Predictors" AND "Postpartum Mothers" AND ‘‘Ethiopia’’
	#8	Limit to February 18, 2022
	#9	Only Humans
	#10	Only English language
	#11	"Completion" OR "Completing" AND "maternity continuum of care" AND "Associated factors" OR "Determinates factors" OR "Predictors" AND "Postpartum Mothers" AND "Limit to February 18, 2022" AND "Only Humans" AND "Only English language"
Scopus, Google scholar, African Journals Online, and Web of Sciences	#1	"Completion" OR "Completing" AND "maternity continuum of care" AND "Associated factors" OR "Determinates factors" OR "Predictors" AND "Postpartum Mothers" AND ‘‘Ethiopia’’

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Newcastle-Ottawa Quality Assessment Scale scale for cross sectional studies	**Selection**				**Comparability**	**Outcome**		**Total score**
	Representativeness (1)	Sample size (1)	Non-respondents (1)	Ascertainment of the exposure (risk factor) (2)	The subjects in different outcome groups are comparable, based on the study design or analysis. confounding factors are controlled (2)	Assessment of the outcome (2)	Statistical test (1)	
Shitie A. et al (15)	1	1	1	1	1	1	1	7
Tizazu MA. et al (14)	1	1	1	1	1	2	1	7
Cherie N.et al(18)	1	1	1	1	1	2	1	8
Asratie MH. et al(20)	1	1	1	2	1	2	1	8
Haile D. et al(17)	1	1	1	1	1	2	1	7
Atnafu A. et al(19)	1	1	1	2	1	2	1	7
Emiru AA. et al (10)	1	1	1	2	1	2	1	8
Sertsewold SG. Et al (16)	1	1	1	1	1	2	1	7
Tsega D. et al (25)	1	1	1	1	1	1	1	7

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Factors associated with completion of maternity continuum of care in Ethiopia: A systematic review and meta-analysis

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Factors associated with completion of maternity continuum of care in Ethiopia: A systematic review and meta-analysis

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Abstract

Objective The present study aimed to estimate the pooled prevalence of completion of the maternity continuum of care and its associated factors in Ethiopia.

Study design Systematic review and meta-analysis.

Study setting Ethiopia.

Study participants A total of 6,245 reproductive-age women were included.

Primary outcome The pooled prevalence of completion of the maternity continuum of care.

Secondary outcome Factors associated with completion of the maternity continuum of care.

Methods We systematically searched international databases such as PubMed, Scopus, African Journals Online, Google Scholar, and Web of Sciences to retrieve related articles. The PRISMA guidelines were used for this study. Publication bias was assessed using the Funnel plot and Eggers test. Evidence of heterogeneity was checked using Cochrane Q-test and I^2 statistics. Subgroup analysis was computed based on the study regions. Data were analyzed using STATA version 14 statistical software. Weighted inverse variance random effect models were used to estimate the pooled prevalence of completion of the maternity continuum of care.

Results The pooled prevalence of completion of maternity continuum care in Ethiopia was 25.82% (95%CI; 16.69%-34.94%). Urban residence (AOR=2.77, 95%CI; 1.99-3.86), having secondary and above educational status (AOR=3.50, 95%CI; 2.50-50) pre-pregnancy contraceptive utilization (AOR=3.25, 95%CI; 2.02-5.22), women's autonomy (AOR=3.81, 95%CI; 2.74-5.31), following mass media (AOR=2.51, 95%CI; 1.79-3.50), early initiation of ANC (AOR=4.98, 95%CI; 3.28-7.57), planned pregnancy (AOR=2.93, 95%CI; 1.99-4.32), birth preparedness and complication readiness (AOR=1.80, 95%CI; 1.29-2.51), distance from a health facility <30 minutes (AOR=3.29, 95%CI; 2.45-4.42) were factors associated with completion of maternity continuum care in Ethiopia.

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Conclusion The pooled prevalence of completion of maternity continuum care in Ethiopia was low. Therefore, policymakers and stakeholders should improve the completion of antenatal care, the rate of skilled birth attendants, and postnatal follow-up. Enhancing the accessibility of health facilities, women's awareness, and empowering women's decision-making are recommended.

Key words: Completion, Continuum of care, Antenatal care, Skilled birth attendants, Postnatal follow-up, Ethiopia

Strength and limitation of study

- To our knowledge, this is the first study conducted in Ethiopia to estimate the pooled prevalence of completion of the maternity continuum of care and its associated factors.
- This study gives recent information on the factors associated with the completion of the maternity continuum of care.
- We strictly followed PRISMA guidelines and assessed the quality of each study using the modified Newcastle Ottawa scale.
- The absence of study in some regions of Ethiopia might make it unable to generalize the finding of this study.
- Variations between studies in assessing the completion of antenatal care and postnatal follow-up (i.e. within two days or 6 weeks) might affect the outcome variables of this study. However, each included study assessed all components of the maternity continuum of care.

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Background

Globally, maternal and neonatal mortality is overwhelming [1]. Approximately, 295,000 women died due to preventable pregnancy and childbirth-related complications in 2017. Of these, 94% of death occurred in low-resource seating. Sub-Saharan Africa and Southern Asia accounted for the highest maternal mortality [2]. In Ethiopia, maternal mortality is 412 per 10000 live births, and neonatal mortality is 30 per 1000 live births [3]. Ethiopia has planned to reduce maternal mortality below or fewer than 70 per 100,000 live births by the end of the 2030 Sustainable Development Goals [4].

Continuum of care (CoC) is a crucial strategy in reducing maternal and neonatal mortality in low and middle-income countries [5,6]. CoC has two components. Firstly, the sequential time component includes antenatal care, skilled birth attendants, and postnatal follow-up. Secondly, the space component includes the provision of care at the community-family level and health facility [7]. Evidence shows that CoC could avert half a million maternal deaths, 4 million neonatal deaths, and 6 million child death [8,9]. However, completion of the maternity continuum of care was a challenge in low-income countries [10]. Studies conducted in Tanzania, Ghana, and Nepal showed that only 8% to 41% of women receive complete maternity CoC [11–13].

Completion of the maternity continuum care and its associated factors were studied in Ethiopia. For instance, studies conducted in the Southern region reported that only 9.7% of mothers complete the maternity continuum of care while in the Amhara region 12.1% to 45% of mothers completed the maternity continuum of care [10,14–20]. Evidence revealed that timely initiation of antenatal care before 16 weeks, birth preparedness and complication readiness, educational status, discussion with husband, and being satisfied with the service

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delivery were some of the factors associated with completion of the maternity continuum of care [15,17,18,21].

Inconsistent findings were reported regarding the completion of the maternity continuum of care and its associated factors across different regions of Ethiopia. For instance, a study conducted by Shitie A. et al in the Amhara region reported that 42% of women had a complete maternity continuum of care [15]. However, a study conducted by Haile D, et al in the Southern region reported that only 9.7% of women had a complete maternity continuum of care [17]. Additionally, there are no nationally representative pooled data on the prevalence of completion of the maternity continuum of care in Ethiopia.

Accordingly, understanding the overall prevalence of completion of the maternity continuum of care (i.e. ANC 4+, SBA, and at least one PNC following delivery) and identifying its associated factors is crucial for policymakers, governments, and stakeholders. Therefore, the main objective of this study was to estimate the pooled prevalence and to identify factors associated with the completion of the maternity continuum of care in Ethiopia.

Methods

Study design and setting

According to the World Bank, Ethiopia is one of the low-income countries located in the horn of Africa with a total population of 115 million, and by the end of 2060; it is projected to be 210 million. A systematic review and meta-analysis was conducted to estimate the pooled prevalence of completion of the maternity continuum of care and its associated factors in Ethiopia. Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines were used for this review (Supplementary File 1) [22].

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Information Sources

We have checked the PROSPERO database (http://www.library.ucsf.edu/) to rule out whether published or on-going projects exist related to the completion of the maternity continuum of care to avoid further duplication. Thus, the finding revealed that there were no on-going or published articles in the area of this topic. Therefore, the protocol of this review is registered at PROSPERO databases with ID=CRD42022312692. International databases such as PubMed, Google Scholar, Scopus, African Journals Online, and Web of Sciences were searched to retrieve related articles. Grey literature was searched using Google. Search terms were formulated using PICO guidelines through online databases. Medical Subject Headings (MeSH) and key terms had been developed using different Boolean operators ''AND'' and ''OR'' (Supplementary File 2).

Eligibility criteria

The inclusion criteria were studies which report the overall prevalence of completion of the maternity continuum of care and its associated factors. Both published and unpublished including pre-print studies at any time in the English language only were considered. Lastly, all community-based cross-sectional study designs conducted in Ethiopia till the last search day February 18, 2022, were included. Regarding the study period, there was no restriction. The exclusion criteria were those articles without full abstracts or texts and articles reported out of the outcome interest. Moreover, citations without abstracts and/or full-text, anonymous reports, commentaries, letters, editorials, reviews, systematic reviews, and meta-analysis were excluded after reviewing the full texts.

Outcome measurements

The outcome variables of interest of the study were the uptake of the CoC at antepartum, intrapartum, and postpartum stages: 1) continuum of care at the antepartum stage is women

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who received four or more antenatal care (ANC 4+) visits, 2) continuum of care at the intrapartum stage is those women who continued use of skilled birth attendance after receiving ANC 4+ visits, and 3) continuum of care at the postpartum stage or complete continuum is those women who received PNC for the mothers and their newborns, within six weeks of their delivery after receiving both ANC 4+ visits and delivered by skilled assistance [7,8]. The secondary aim was to identify factors associated with the completion of the maternity continuum of care.

Data extraction

All relevant studies were exported to Endnote version X8 software. Duplicated studies were removed and exported to a Microsoft Excel spreadsheet. Two authors (AM and KH) independently extracted all the important data using a standardized data extraction form which was adapted from Joanna Briggs Institute (JBI) [23]. For the primary outcome (prevalence) the data extraction format included (the primary author, publication year, study period, study setting, objective of the study, sampling procedure, mode of interview, region, sample size, prevalence, and study design). Similarly, two authors (AM and AT) extracted data for the secondary outcome (factors associated with completion of the maternity continuum of care) using a Microsoft Excel spreadsheet (i.e. adjusted odds ratio, lower confidence interval, and upper confidence interval). Finally, each factor was exported into STATA version 14 software to identify factors associated with completion of the maternity continuum of care.

Quality and risk of bias assessment

Three authors (AM, KH, and AT) assessed the quality of each study using the modified Newcastle Ottawa Quality Assessment Scale (NOS) for cross-sectional studies [24]. The NOS had a maximum of 9 points. It contains sample selection, comparability and outcome

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measurement; accordingly four points, two points, and three points were given respectively. The study which scored more than 7 points was included (Supplementary File 3).

Data processing and analysis

Firstly, selected articles were entered into Microsoft Excel spread sheet format and imported to STATA version 14.0 Statistical software for analysis. A narrative synthesis was used to analyse and interpret the findings. A random-effects meta-analysis model was used to pool the estimates of the magnitude of COC and its associated factors accounting for the variability among studies. Cochrane Q-test and I^2 statistics were computed to assess heterogeneity among all studies. Accordingly, if the result of I^2 is 0% to 40% it is mild heterogeneity, 30 to 60% would be moderate heterogeneity, 50 to 90% would be substantial heterogeneity; and 75 to 100% would be considerable heterogeneity. Funnel plot and Eggers test were done to assess publication bias. The p value >0.05 indicated that there was no publication bias. Subgroup analysis was done based on the study region. Forest plot format was used to present the pooled prevalence completion of the maternity continuum of care with 95%CI. To identify factors associated with completion of the maternity continuum of care, STATA version 14.0 Software was used.

Public and patient involvement

Neither patients nor public were involved in this study.

Results

Characteristics of included studies

Overall the search strategy yielded 233 articles. After removing duplicated files, there were 102 articles remaining. After reviewing, (n=63) articles were excluded by title, and (n=21) articles were excluded by reading abstracts. Therefore, 18 full-text articles were accessed and assessed for inclusion criteria, resulting in the further exclusion of 9 articles primarily due to

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reasons (Figure 1). As a result, 9 studies were fulfilled the inclusion criteria and undergo the final meta-analysis (Table 1)

Table 1: Study characteristics included in the systematic review and meta-analysis on the prevalence of completion of maternity continuum of care and its associated factors in Ethiopia.

Study Id	Authors	Publication year	Study period	Study setting	Objective of the study	Sampling procedure	Mode of interview	Region	Sample size	Prevalence	Study design	NOS
1	Shitie A. et al [15]	2020	February 25 to March 10, 2019	Enemay district	Assess completion Maternity Continuum of Care Identify Factors Associated with	simple random sampling technique	face-to-face interview	Amhara	651	45%	CS	7
2	Tizazu MA. et al [14]	2020	February 17 to March 15,2020	Debre Berhan town	utilization of continuum maternity care Identify associated factors	cluster sampling technique		Amhara	647	37.2%	CS	7
3	Cherie N.et al [18]	2021	Feb-Mar 2020	Legambo district	Maternity continuum of care Determinant factors	Multistage sampling technique	face-to-face interviews	Amhara	732	11.2%	CS	8
4	Asratie MH. et al [20]	2020	March 12, 2019 to April1, 2019	Motta town and Hulet Eji Enese district	Assess completion of maternity continuum of care Identify associated factors	stratified cluster sampling technique	face to face interviews	Amhara	819	47%	CS	8

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5	Haile D. et al [17]	2020	15 February to 15 March 2019.	Arba Minch Health and Demographic Surveillance System (HDSS) site	Level of completion along continuum of care Identifying associated factors	simple random sampling technique	face-to-face interviews	SNNP	432	9.7%	CS	7
6	Atnafu A. et al [19]	2020	May 01 to June 29, 2019	Dabat and Gondar Zuria rural Districts	Completion of the continuum of maternal healthcare services. Identify associated factors	simple random sampling technique	face-to-face interviews	Amhara	565	21.6%	CS	7
7	Emiru AA. et al [10]	2020	June 2018	West Gojjam Zone	Completed the continuum of maternal care services. Factors affecting retention on the continuum of care	two-stage cluster sampling	face-to-face interviews	Amhara	1281	12.1%	CS	8
8	Sertsewold SG. et al [16]	2021	February 2020 to 10 March 2020	Siyadebirena Wayu district	Utilisation of continuum of maternal healthcare services. Identify associated factors	simple random sampling	face-to-face interviews	Amhara	614	16.1%	CS	7

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9	Tsega D. et al [25]	2022	March 10 to March 30, 2020	Ebinat district	Maternity continuum care completion. Identifying associated factors	simple random sampling technique	face-to-face interviews	Amhara	504	37.6%	CS	7

NB: NOS; Newcastle Ottawa Quality Assessment Scale; CS, Cross Sectional

Completion of the maternity continuum care in Ethiopia

The pooled prevalence of completion of the maternity continuum of care in Ethiopia was 25.82% (95%CI; 16.69%-34.94%). The result of Cochrane heterogeneity index is (I^2=98.9%), P=0.000, indicated that there is substantial heterogeneity of different studies (I^2>50%). The finding was presented using a forest plot (Figure 2).

Publication bias

A funnel plot was done to check the presence of publication bias at a significance level of less than 0.05. The Egger's regression test was not statistically significant (p>0.05) confirming no evidence of publication bias, as presented by the funnel plot (Figure 3).

Subgroup analysis of completion of the maternity continuum care

The finding of subgroup analysis by region showed that the pooled prevalence of completion of the maternity continuum of care was highest in the Amhara region 21.53% (95% CI: 20.52-22.53), I^2 =98.9%, p <0.001) and lowest in the SNNP region 9.72% (95% CI: 6.93-12.52), I^2 = 98.9%, p <0.001) (Figure 4).

Sensitivity analysis

A random effect model result showed that no single study has influenced the pooled prevalence of completion of the maternity continuum care (Table 2).

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Table 2: Sensitivity analysis of studies included in the systematic review and meta-analysis on the prevalence of completion of maternity continuum care in Ethiopia.

Study omitted	Estimate(95% CI)
Shitie A. et al (2020)	16.07(6.07-42.55)
Tizazu MA. et al (2020)	16.18(6.10-42.90)
Cherie N.et al (2021)	18.61(6.61-52.43)
Asratie MH. et al (2020)	15.69(5.88-41.86)
Haile D. et al (2020)	18.35(6.74-49.99)
Atnafu A. et al (2020)	16.77(6.28- 44.79)
Emiru AA. et al (2020)	19.5(6.42-59.22)
Sertsewold SG. et al (2021)	17.18(6.34-46.59)
Tsega D. et al (2022)	16.42(6.24-43.24)
Combined	17.07(6.65- 43.84)

Factors associated with completion of the maternity continuum of care

In this systematic review and meta-analysis; urban residence, secondary and above educational status, pre-pregnancy contraceptive utilization, autonomy, following mass media, early initiation of ANC, planned pregnancy, birth preparedness and complication readiness, and distance from health facility less than 30 minutes were factors associated with completion of the maternity continuum of care in Ethiopia.

The odds of being residing in an urban area were 2.77 times more likely to complete the maternity continuum of care than those residing in a rural area (AOR=2.77, 95%CI; 1.99-3.86). Those women who had secondary and above educational status were 3.5 times more likely to complete the maternity continuum of care than those who had no formal education (AOR=3.5, 95%CI; 2.5-5). The odds of utilizing pre-pregnancy contraceptives were 3.25 times more likely to complete the maternity continuum of care than those who had not utilized contraceptive methods (AOR=3.25, 95%CI; 2.02-5.22). The odds of being autonomous in health care decisions were 3.81 times more likely to complete the maternity

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continuum of care than their counterparts (AOR=3.81, 95%CI; 2.74-5.31). The odds of following mass media were 2.51 times more likely to complete the maternity continuum of care than those women who did not follow mass media (AOR=2.51, 95%CI; 1.79-3.50). Those women who had early initiation of ANC were 4.98 times more likely to complete the maternity continuum of care than those women who had late initiation of ANC (AOR=4.98, 95%CI; 3.28-7.57). Those women who had planned pregnancy were 2.93 times more likely to complete the maternity continuum of care than those who had unplanned pregnancy (AOR=2.93, 95%CI; 1.99-4.32). The odds of having birth preparedness and complication readiness were 1.8 times more likely to complete the maternity continuum of care than their counterparts (AOR=1.80, 95%CI; 1.29-2.51). Furthermore, women who found less than 30 minutes distance from a health facility were 3.29 times more likely to complete the maternity continuum of care than their counterparts (AOR=3.29, 95%CI; 2.45-4.42) (Table 3).

Table 3: Factors associated with completion of maternity continuum of care in Ethiopia: a systematic review and meta-analysis, 2022.

Variables	Included studies	Odds Ratio(95%CI)	Pooled odds ratio(95%CI)	Heterogeneity (I^2 ,p-value)
Urban residence	Sertsewold SG. et al Tsega D. et al Cherie N.et al Asratie MH. et al	3.91(1.06-14.39) 2.45(1.35-4.45) 1.84(1.03-3.29) 4.60(2.50-8.50)	2.77 (1.99, 3.86)	39.7%, 0.173
Secondary and above educational status	Shitie A. et al Sertsewold SG. et al Tsega D. et al Tizazu MA. et al Asratie MH. et al	6.00(2.26-16.60) 5.36(1.15-25.06) 2.75(1.42-5.32) 3.11(1.32-7.31) 3.50(1.90-6.30)	3.50(2.45, 5.00)	0.0%, 0.735
Pre-pregnancy contraceptive utilization	Haile D. et al Cherie N.et al Sertsewold SG. et al	3.90(1.40-11.00) 2.70(1.50-5.04) 4.95(1.61-15.20)	3.25 (2.02,5.22)	0.0%, 0.601
Women autonomy	Shitie A. et al Sertsewold SG. et al Cherie N.et al Asratie MH. et al	4.00(2.26-7.20) 4.45(1.69-11.60) 3.70(1.90-7.20) 3.50(1.90-6.30)	3.81 (2.74,5.31)	0.0%, 0.976
Following mass media	Shitie A. et al Tsega D. et al	1.97(1.20-3.27) 3.64(2.02-6.56)	2.50 (1.79,3.50)	19.0%, 0.291

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	Cherie N.et al	2.33(1.15-4.74)		
Early initiation of ANC	Tizazu MA. et al Haile D. et al Emiru AA. et al	2.57(1.41-4.68) 10.70(5.10-22.70) 7.53(2.94-19.29)	4.98 (3.28, 7.57)	78.8%, 0.009
Planned pregnancy	Shitie A. et al Cherie N.et al Haile D. et al	3.33(1.87-5.90) 2.45(1.36-4.40) 3.50(1.10-11.40)	2.93 (1.99,4.32)	0.0%, 0.728
Birth preparedness and complication readiness	Haile D. et al Tizazu MA. et al	2.90(1.40-6.10) 1.59(1.10-2.32)	1.80 (1.29,2.51)	50.9%, 0.153
Distance from health facility(<30 minutes)	Sertsewold SG. et al Tsega D. et al Atnafu A. et al Asratie MH. et al	3.17(1.38-7.25) 3.22(1.84-5.63) 4.98(2.97-8.38) 2.10(1.20-3.70)	3.29(2.45,4.42)	38.9%, 0.178

Discussion

The current study revealed that the estimated prevalence of completion of the maternity continuum of care was 25.82%. Urban residence, secondary and above educational status, pre-pregnancy contraceptive utilization, autonomy, following mass media, early initiation of ANC, planned pregnancy, birth preparedness and complication readiness, and distance from health facility <30 minutes were factors associated with completion of the maternity continuum of care in Ethiopia.

The prevalence of completion of the maternity continuum of care was low in Ethiopia. The finding is higher than a study conducted in Ghana (8%) [12]. The possible explanation might be due to the study period difference. For instance, this study included recent studies as compared to a study done in Ghana which was conducted from 2011 to 2013. However, the finding is lower than studies conducted in Zambia (38%), Cambodia (60%), and Egypt (50.4%) [26–28]. The possible explanation for varieties of findings might be due to differences in accessing healthcare facilities. For example, a study conducted in Cambodia might have better access to quality maternity care and can easily access healthcare services.

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Women who resided in urban areas were 2.77 times more likely to complete the maternity continuum of care than those who resided in rural areas. The possible explanation could be those women who resided in rural areas might not easily access transportation services, particularly during night time, and there might be a lack of access to a health facility in their nearby residence. Thus, the probability of discontinuing the maternity continuum of care is more likely than those women who resided in urban areas. The finding is in line with a study conducted in Ghana [12].

Those women who had completed secondary and above educational status were 3.5 times more likely to complete the maternity continuum of care than those who did not have formal education. The finding is consistent with studies conducted in Pakistan, Uganda, and Kenya [29,30]. The possible explanation might be those women who are able to read and write have awareness regarding the benefit of antenatal follow-up, giving birth at a health facility, and postnatal follow-up such as early detection and management of diseases, prevention of infection, and bleeding after delivery.

Those women who had utilized pre-pregnancy contraceptive methods were 3.25 times more likely to complete the maternity continuum of care than their counterparts. The possible explanation might be the integration of the family planning service with other maternal and child healthcare services in the health facility. Therefore, women who had utilized contraceptive methods have an awareness concerning the maternity continuity of care and that might enable them to continue antenatal care, childbirth, and postnatal follow-up at a health facility [31].

Women who had autonomy in healthcare decision-making were 3.8 times more likely to complete the maternity continuum of care than those who did not have autonomy. The finding is in line with a study conducted in Gambia [32]. This could be explained that

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autonomous women have the ability to decide regarding their sexual and reproductive health service needs. The finding is supported by a review conducted in developing countries showed that women's autonomy is crucial to continue reproductive and child healthcare services [33].

The odds of following mass media were 2.5 times more likely to complete the maternity continuum of care than those who did not follow mass media. The finding is similar to previous studies conducted in Zambia and Sub-Saharan Africa [27,34]. The possible explanation might be media have a significant role in creating awareness on the maternity continuum of care.

Those women who had initiated early antenatal care before 16 weeks were 4.98 times more likely to complete the maternity continuum of care than their counterparts. The possible explanation might be antenatal care is the entry into the maternity continuum of care. Therefore, women who had started early ANC visits receive counselling on the continuum of care that might enable them to complete the maternity continuity care. The finding is supported by studies conducted in India, Tanzania, and Ethiopia indicating that early ANC visit could enhance the maternity continuum of care [13,35,36]. On the other hand, a study conducted in India and Sub Saharan Africa shows that low utilization of ANC service was a major obstacle to the maternity continuum of care [37, 38].

Those women who had planned pregnancies were 2.93 times more likely to complete the maternity continuum of care than those who had unplanned pregnancies. This could be explained that women who had unplanned pregnancies might have fear of visiting healthcare facilities. The finding is consistent with a study conducted in Ghana [12]. Similarly, those women who had birth preparedness and complication readiness were 1.8 times more likely to complete the maternity continuum of care than their counterparts. This could be explained

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that those women who had birth preparedness and complication readiness have awareness about the maternity continuum of care. Moreover, women who found < 30 minutes distance from the health facility were 3.29 times more likely to complete the maternity continuum of care than their counterparts. The finding is in line with a study conducted in Lao PDR and Nigeria [21,39]. The possible explanation might be as the health facility is far away from their home; there might be a lack of transportation services.

This study has several strengths. To our knowledge, this systematic review and meta-analysis is the first study conducted in Ethiopia to estimate the pooled prevalence of completion of the maternity continuum of care and its associated factors. Additionally, this study gives recent information on the factors associated with the completion of the maternity continuum of care. Moreover, we strictly followed PRISMA guidelines and assessed the quality of each study using the modified Newcastle Ottawa scale. However, this study is not void of limitations. Variations between studies in assessing the completion of antenatal care and postnatal follow-up (i.e. within two days or 6 weeks) might affect the outcome variables of this study. Moreover, the absence of study in some regions of Ethiopia (i.e. Afar, Gambella, Somalia, and Tigray region) might make it unable to generalize the finding of this study. Thus, we recommend that future studies should be conducted in the above respective regions.

Conclusion

The pooled prevalence of completion of the maternity continuum of care was low in Ethiopia. Urban residence, secondary and above educational status, pre-pregnancy contraceptive utilization, women's autonomy, following mass media, early initiation of ANC, planned pregnancy, birth preparedness and complication readiness, and distance from health facility < 30 minutes were factors associated with completion of the maternity continuum of care in Ethiopia. Therefore, policymakers and stakeholders should improve the completion of

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antenatal care, the rate of skilled birth attendants, and postnatal follow-up. Enhancing the accessibility of health facilities, women's awareness, and empowering women's decision-making are recommended.

Abbreviations

ANC; Antenatal Care; CoC; Continuum of Care; SBA; Skilled Birth Attendant; PNC; Postnatal Care; NOS; Newcastle Ottawa Quality Assessment Scale; WHO; World Health Organization, PRISMA; Preferred Reporting Items for Systematic Reviews and Meta-Analyses

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The data set analysed during the current study is available from the cross-ponding author upon reasonable request.

Competing Interests

All authors declare that they have no competing interests.

Funding

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Authors' contributions

AM conceived the idea and participated in data extraction, analysis, and draft writing. AM, KH, and AT participated in the analysis, preparation of the manuscript, and revision. All

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authors read and approved the final version of the manuscript to be considered for publication.

Institutional Review Board Statement:

Not applicable. This systematic review and meta-analysis used and analysed information obtained from pre-existing studies.

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Legends

Figure 1; Flow chart of selection for systematic review and meta-analysis on completion of maternity care and its associated factors in Ethiopia, 2022

Figure 2; Pooled prevalence of completion of maternity care in Ethiopia, 2022

Figure 3; Funnel plot showing symmetric distribution of articles on completion of maternity continuum of care in Ethiopia, 2022

Figure 4; Forest plot showing Sub group analysis of completion of maternity continuum of care and its associated factors by region in Ethiopia, 2022

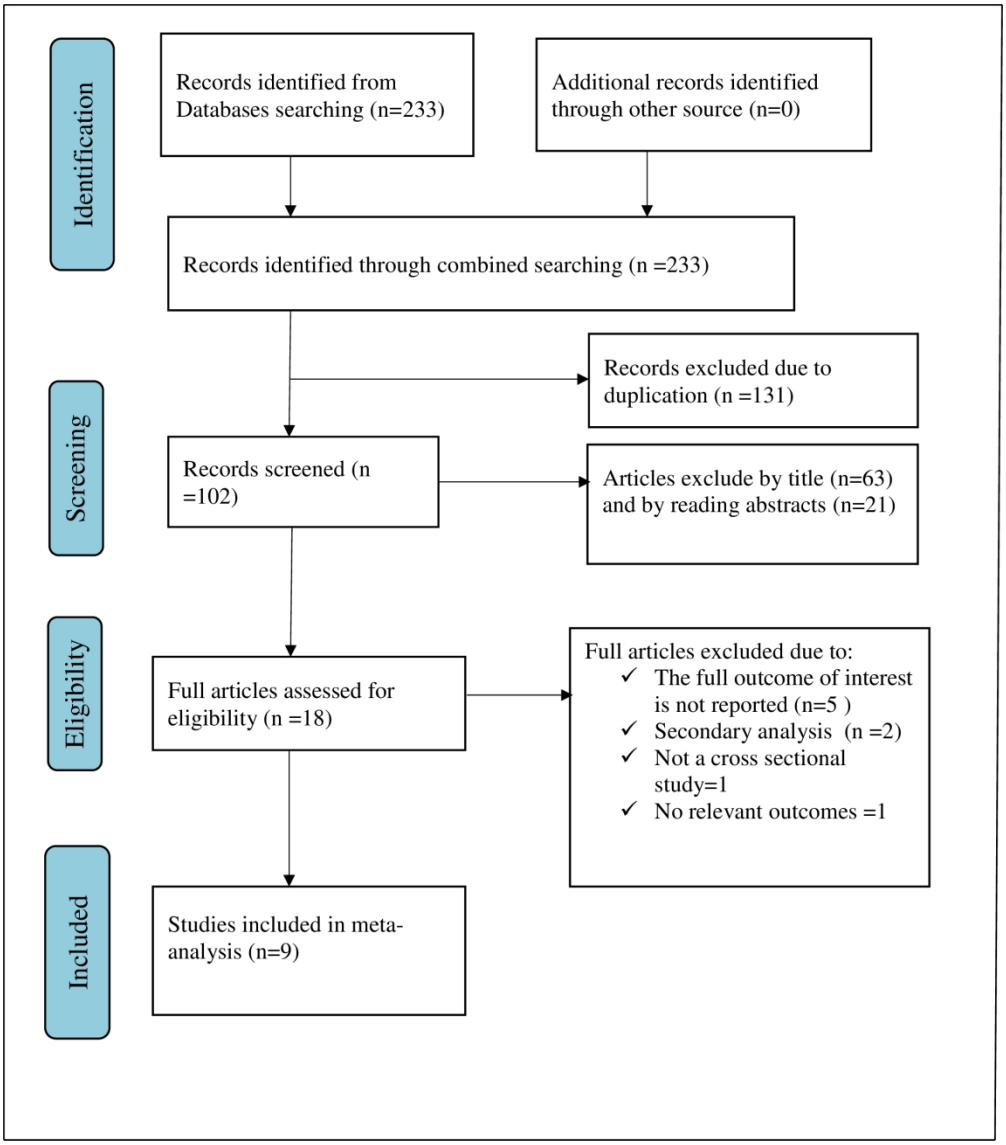
Supporting information

Supplementary File 1: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Guideline.

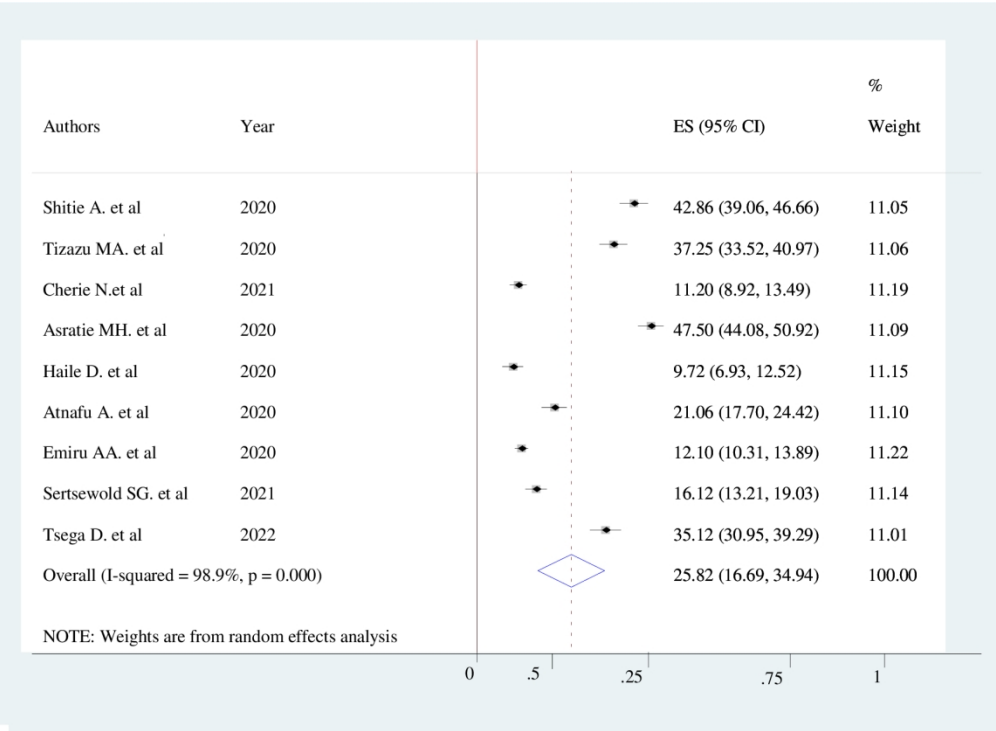
Supplementary File 2: Search string for databases.

Supplementary File 3: Newcastle-Ottawa Quality Assessment Scale of 9 cross sectional studies included in the review of the status of completion of maternity continuum of care in Ethiopia

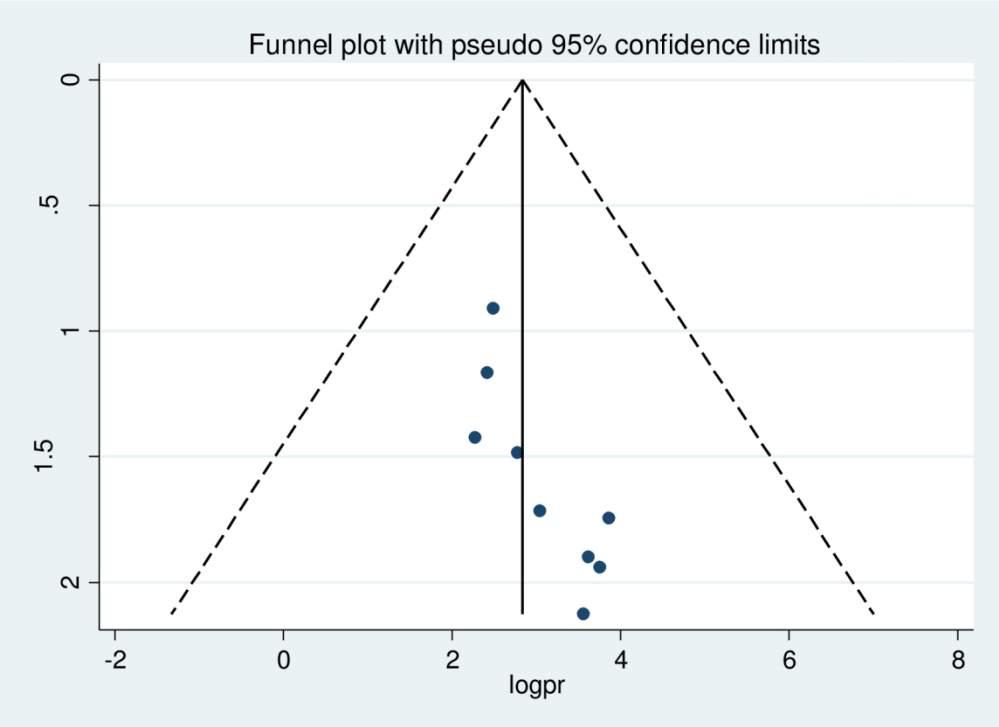
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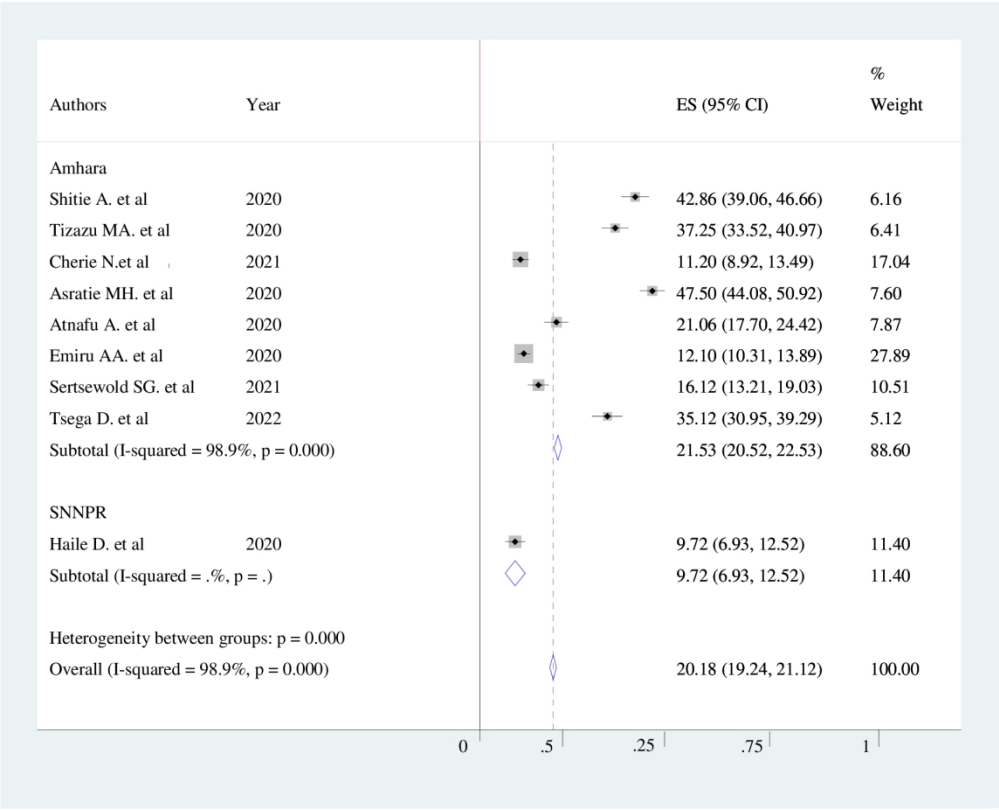
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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2 to 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 3
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 5 to 6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 7
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 7
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 6
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 7
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 7
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 7
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 9
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 9
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 9
Study characteristics	17	Cite each included study and present its characteristics.	Page 8
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 8 to page 13
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Page 8 to page 13
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page 8 to page 13
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Page 8 to page 13
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 8 to page 13
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Page 8 to page 13
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 8 to page 13
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 8 to page 13
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 14 to 17
	23b	Discuss any limitations of the evidence included in the review.	Page 18
	23c	Discuss any limitations of the review processes used.	Page1 8
	23d	Discuss implications of the results for practice, policy, and future research.	Page 18
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 5 to 6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 5
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 13
Competing interests	26	Declare any competing interests of review authors.	Page 12
Availability of data, code and	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 18 to 19

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
other materials			

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

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Databases	Search	Search terms
PubMed	#1	"Completion" OR "Completing"
	#2	"Maternity Continuum of Care"
	#3	"Completion" OR "Completing" AND "maternity continuum of care"
	#4	"Associated factors" OR "Determinates factors" OR "Predictors"
	#5	"Postpartum Mothers"
	#6	"Completion" OR "Completing" AND "maternity continuum of care" AND "Associated factors" OR "Determinates factors" OR "Predictors" AND "Postpartum Mothers"
	#7	"Completion" OR "Completing" AND "maternity continuum of care" AND "Associated factors" OR "Determinates factors" OR "Predictors" AND "Postpartum Mothers" AND ‘‘Ethiopia’’
	#8	Limit to February 18, 2022
	#9	Only Humans
	#10	Only English language
	#11	"Completion" OR "Completing" AND "maternity continuum of care" AND "Associated factors" OR "Determinates factors" OR "Predictors" AND "Postpartum Mothers" AND "Limit to February 18, 2022" AND "Only Humans" AND "Only English language"
Scopus, Google scholar, African Journals Online, and Web of Sciences	#1	"Completion" OR "Completing" AND "maternity continuum of care" AND "Associated factors" OR "Determinates factors" OR "Predictors" AND "Postpartum Mothers" AND ‘‘Ethiopia’’

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Newcastle-Ottawa Quality Assessment Scale scale for cross sectional studies	**Selection**				**Comparability**	**Outcome**		**Total score**
	Representativeness (1)	Sample size (1)	Non-respondents (1)	Ascertainment of the exposure (risk factor) (2)	The subjects in different outcome groups are comparable, based on the study design or analysis. confounding factors are controlled (2)	Assessment of the outcome (2)	Statistical test (1)	
Shitie A. et al (15)	1	1	1	1	1	1	1	7
Tizazu MA. et al (14)	1	1	1	1	1	2	1	7
Cherie N.et al(18)	1	1	1	1	1	2	1	8
Asratie MH. et al(20)	1	1	1	2	1	2	1	8
Haile D. et al(17)	1	1	1	1	1	2	1	7
Atnafu A. et al(19)	1	1	1	2	1	2	1	7
Emiru AA. et al (10)	1	1	1	2	1	2	1	8
Sertsewold SG. Et al (16)	1	1	1	1	1	2	1	7
Tsega D. et al (25)	1	1	1	1	1	1	1	7

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